

11/1/2023

BIG DATA AND BUSINESS INTELLIGENCE

Topic: The Impact of Customer Churn in XYZ Bank

Table of Contents

A. EXECUTIVE SUMMARY	3
Introduction	3
Key Findings	3
Recommendations	4
B. INTRODUCTION	5
i) Data Source	5
ii) BI Questions	7
C. FINDINGS BASED ON ANALYSIS AND EVALUATION	8
Q1 What is the general knowledge of the customer base:	8
Q2 Is there any significance of those that left vs those that remained when compared:	11
Q3 What are contributing to the balance of those that left vs those that remained	16
Q4 What are influencing the credit scores of those that left vs those that remained	19
Q5 Considering some key factors, what is the impact of customer churn rate:	22
D. CONCLUSIONS AND RECOMMENDATIONS	24
Summary	24
Recommendations	25
Personal Conclusion	25
APPENDIX: BI DESIGN	27
I. DATA PRE-PROCESSING AND DATA CLEANING	27
Load and Clean Dataset:	27
Creating Dimensional Tables for Relationships:	28
Merging of Queries:	30
II. DATA MODELLING	31
III. DAX AND M-LANGUAGE	34
i) DAX	34
ii) M-LANGUAGE:	38
IV. DASHBOARD DESIGN	39
Home Page	39
Customer Distribution Page	39
Churn Distribution Page	40
Churn Bivariate Analysis Page	41
Balance Regression Page	41
Credit Score Regression Page	42
AI Insights Page	42
V. REFERENCE	43

VI. SELF-ASSESSMENT TABLE.....43

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A. EXECUTIVE SUMMARY

Introduction

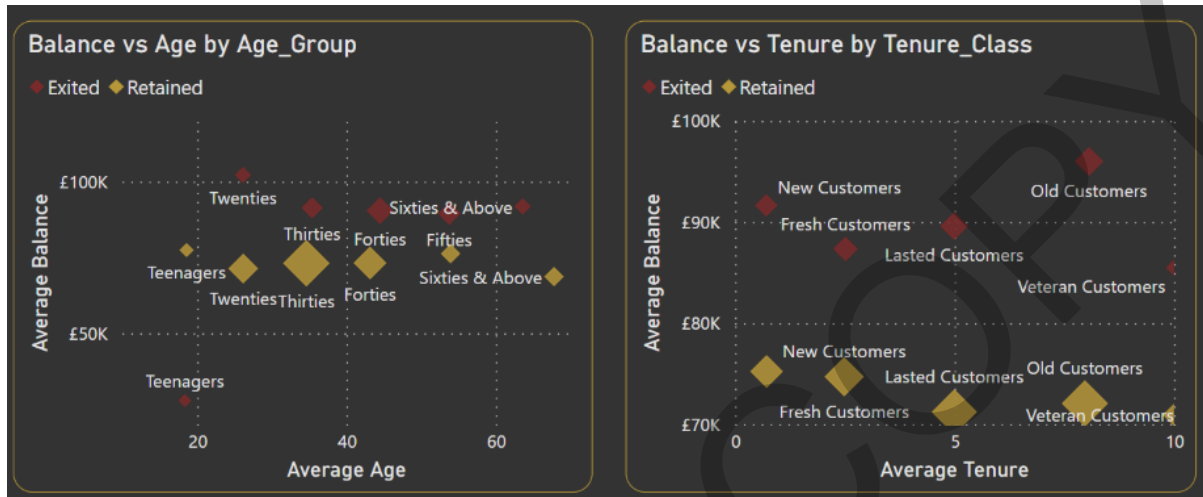
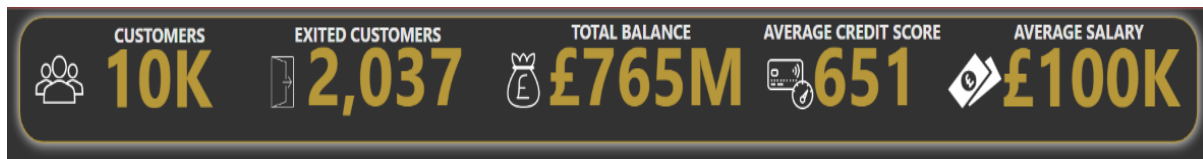
This report is a case study of a bank (XYZ Bank) facing the issue of reducing customer base and needs to find solutions to the emerging crisis. The following were the key questions that have been put up of which this report seeks to analyse and report on:

- ❖ What is the general knowledge of the customer base
- ❖ Is there any significance of those that left vs those that remained when compared
- ❖ What are contributing to the balance of those that left vs those that remained
- ❖ What are influencing the credit score of those that left vs those that remained
- ❖ Considering some key factors, what is the impact of customer churn rate

The dataset used for this case study was gotten from [Kaggle](#) and was selected because it met the requirements of the business questions. 10 tables were created from the original flat table, into a snowflake style of data modelling.

Key Findings

- ❖ There are 10,000 unique customers from the original data source
- ❖ Majority (80%) of the customers are still with the bank, while 20% (2,037) of the overall customer left the bank
- ❖ Approximately 7 in 10 customers have credit card across the customer base, while 30% are not subscribing to it
- ❖ Approximately 60% of customers have stayed more than 3 years since they joined the bank
- ❖ Out of 7k customers with credit card, 1k of them left while 6k are still with bank
- ❖ From 1,873 active zero accounts, 180 of them belong to those that left the bank, while out of 3,105 inactive non-zero accounts, 982 are for the leavers
- ❖ There is a significant correlation between the age and the balance of customers at £168.42 for every increase in age. This shows the more the older people patronize them the more there is chances of harbouring large account balance.
- ❖ There is no correlation between the credit score and the balance of customers
- ❖ In relative comparisons of the churn status, credit card, activity, and gender – to the customer base of the bank, churn status is considered to have most importance. This means that churn rate, whether high or low, has a way of affecting the overall customer base
- ❖ However, the churn rate doesn't affect activeness of customers in the bank
- ❖ By all tenure class, none of the class that left the bank had an average balance below £80k while it's the opposite for the Retained because none of their tenure class passed an average balance of same £80k
- ❖ Overall, considering these factors, it's safe to say the exited customers, though at 20% rate, had impact in the customer base



Recommendations

- ❖ Since there is no impact of the churn rate on the activeness of the customers, the management of the bank should not ensure to convert their inactive accounts to active. Perhaps, those that exited were because of the inactiveness of their accounts
- ❖ The management should build marketing campaigns against zero account balance to their existing customers because there is finding that the tendency of customer to leave increase when their account balance hits zero
- ❖ As there is a positive correlation between age and the account balance, which means older people can get fat figures into the bank, prospective older customers should be targeted
- ❖ It is okay to launch a zero-account opener for new customers since that also increases the customer base. However, strategies should be put in place to grow the accounts
- ❖ If the majority of those that left have credit card – which is more than twice of those that don't, then perhaps the value of the credit card should be revisited or promoted

B. INTRODUCTION

Data Analytics and Business Intelligence, with its importance in Businesses of today cannot be over emphasized. Corporate Businesses of any magnitude, even the Small-Medium Enterprises (SMEs), always fall back to look at the best way to polish and make better business decisions through data.

In business, be it banks, corporate firms, or sales-based companies, there is need to seek answers to questions bordering on business problems like high customer churn rate (attrition), staff reduction, lack of sales, to ameliorate the impacts that can be suffered from it, and possibly put the best practices in place to avoid such occurrence in future.

This report is a case study of a bank (XYZ Bank) facing the issue of reducing customer base and needs to find solutions to the emerging crisis. The following were the key questions that have been put up of which this report seeks to analyse and report on:

- ❖ What is the general knowledge of the customer base
- ❖ Is there any significance of those that left vs those that remained when compared
- ❖ What are contributing to the balance of those that left vs those that remained
- ❖ What are influencing the credit score of those that left vs those that remained
- ❖ Considering some key factors, what is the impact of customer churn rate

i) Data Source

The dataset used for this case study was gotten from [Kaggle](#) by @kmalit - a Student Research Assistant in Germany as at four years ago from the time of this report. The dataset was selected because it met the requirements of the business questions. In addition, it is perfect for a comparative analysis via its binominal columns as per the focus of the solutions.

Categories	Description
File Name	Churn_Modelling.csv
File Size	684.86kB
Total Instances (row)	10,000
Total Fields (column)	14
Numerical Columns	6
Categorical Columns	3
Binominal Columns	3
Key Columns	2

After much pre-processing and data cleaning, data modelling – all of which have been captured in the Appendix section of this report, the dataset was modified into different queries (tables) to help in the free flow of the whole analytics.

Data Description: 10 tables were created from the original flat table, into a snowflake style of data modelling – where 'Churn_Modelling' is the fact table, while 'HasCredit', 'IsActive', 'Churn', 'Gender', 'Geography', 'Age & Tenure', and 'Age_Group' are dimensional tables that are connected directly or indirectly to the main table (fact table); 'Salary Range' and 'Tenure Class' are the two reference tables in the data model which are not connected but related in the data. There are also other calculated columns in some of these tables which are all summarized in the table below:

Table	Column	Description
Churn_Modelling (Fact Table)	Balance	Balance of the customer in the bank
	Credit Score	Credit Score of the customer in the bank
	CustomerID	Unique Identifier of each customer in the bank
	EstimatedSalary	Salary of each employed customer in the bank
	TableIDs	This is general description for all the foreign keys
	Salary_Group (Calculated Column)-referred from Salary Range	This is a calculated column for different salaries based on percentage of the max. Low (0 - 33.33% of the maximum salary), Mid (33.33 – 66.67%), High (above 66.67%)
	Other Calculated Columns	This is the general description of other calculated columns based on regression in the analytics
HasCredit (1:* relationship)	CreditCard	Yes – has credit card, No – doesn't have credit card
	HasCreditID	Originally from the dataset turned to Table Key – where 1 = Yes and 0 = No
IsActive (1:* relationship)	Activity	Shows customers that are active or inactive
	IsActiveID	Originally from the dataset turned to Table Key – where 1 = Active and 0 = Inactive
Churn (1:* relationship)	ChurnStatus	Exited – left the bank, Retained – remained in bank
	ChurnID	Originally from the dataset turned to Table Key – where 1 = Exited and 0 = Retained
Gender (1:* relationship)	Gender	Gender of the customer
	GenderID	Table key generated from index column – where 1= Male and 0 = Female
Geography (1:* relationship)	Geography	Resident Country of the customer
	GeographyID	Table key generated from index column
Age & Tenure (1:1 relationship) <i>also the bridge table between Age_Group and the Fact Table</i>	Age	Age of the customer
	Age_GroupID	Foreign key to another table that is connected to it
	Customer_Tenure	Number of years since customer joined the bank
	CustomerID	ID used to represent age and tenure of each customer
	Tenure Class	Two calculated columns referred from the Tenure Class table - based on ranges of customer tenure
Age_Group (1:* relationship) <i>Directly connected to the Age and Tenure but indirectly connected to fact</i>	Tenure Group	
	Age_Bracket	Age bracket of the customer created based on the age of the customer
	Age_Group	Age group of the customer based on the age bracket
	Age_GroupID	Table key generated from the index column
	Lower_Age Upper_Age	The two columns are simple lower and upper age limit for each age bracket
Salary Range (Reference Tale)	Salary_Bracket	Salary bracket of the customer created based on the salary of the customer
	Salary_Range	Salary range of the customer based on the salary bracket
	Lower_Salary Upper_Salary	The two columns are simple lower and upper salary limit for each salary bracket

Tenure_Class (Reference Table)	Tenure_Bracket	Tenure bracket of the customer created based on the tenure of the customer
	Tenure_Group	Tenure group of the customer based on the tenure bracket
	Lower_Tenure	The two columns are simple lower and upper tenure limit for each tenure bracket
	Upper_Tenure	

ii) BI Questions

1. What is the general knowledge of the customer base:
 - a. Distribution across gender, age group, geography
 - b. Customer Engagements: credit card status, active status, churn status, tenure
 - c. Financial Status: tone of deposits (balance), income (salary), credit score
2. Is there any significance of those that left vs those that remained when compared:
 - a. Churn distribution across age group, gender, geography
 - b. Churn distribution across engagements: credit card, activity, accounts, tenure
 - c. Churn comparisons across the financial status: balance, salary, credit score
 - d. How does the financial engagements cut across other dimensions: age group, tenure class
3. What are contributing to the balance of those that left vs those that remained
 - a. Does the balance of those that left depend on their estimated salaries
 - b. How does age affect the balance of those that left the bank
 - c. Is there a direct correlation between tenures of those that left and their balance
4. What are influencing the credit score of those that left vs those that remained
 - a. How does balance affect the credit score of those that left the bank
 - b. Is there a direct correlation between tenures of those that left and their credit scores
 - c. Does the credit scores of those that left depend on their estimated salaries
5. Considering some key factors, what is the impact of customer churn rate:
 - a. Does it affect the customer base
 - b. Does it affect the current balance in the bank
 - c. Does it reduce the activeness of customers

C. FINDINGS BASED ON ANALYSIS AND EVALUATION

Q1 What is the general knowledge of the customer base:

There are 10,000 customers as we have seen from the data source – represented by unique customer ID. This was also calculated by counting each of these different IDs using DAX measure: Customers = DISTINCTCOUNT(Churn_Modelling[CustomerId]).

a. Distribution of Customers across gender, age group, and geography

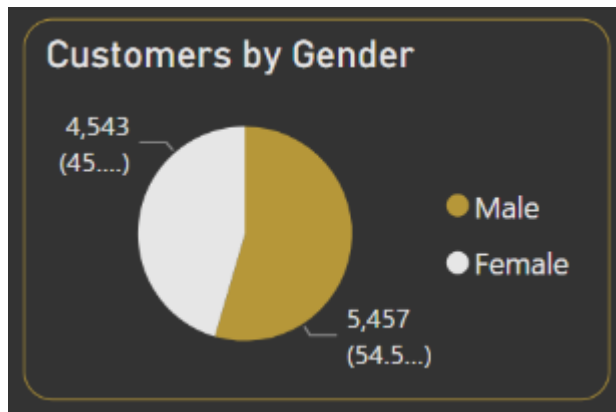


Figure 1: Customer by Gender

Title	Customer Distribution by Gender
Chart Type	Pie Chart
Reason	To give the distribution of the customer base in terms of number and percentage of male and female
Findings	There are more Males than Females according to the chart with about 10% gap

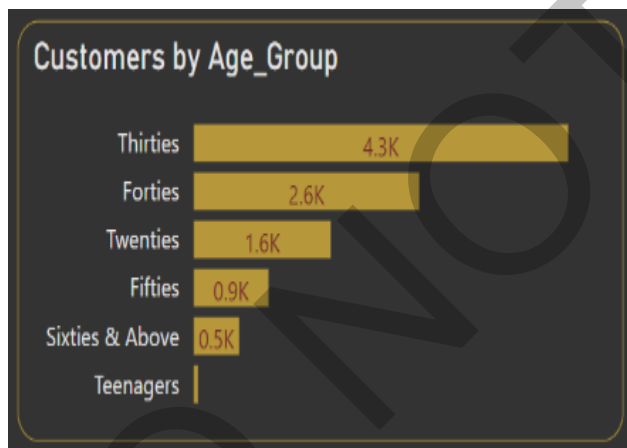


Figure 2: Customer by Age_Group

Title	Customer by Age_Group
Chart Type	Horizontal Bar Chart
Reason	To give the distribution of customer base (quantity) across different age groups (categorical) using lengths of the bars in comparison
Findings	Almost half of the base is in their thirties, while Forties are slightly above 25% base. Teenagers are the least

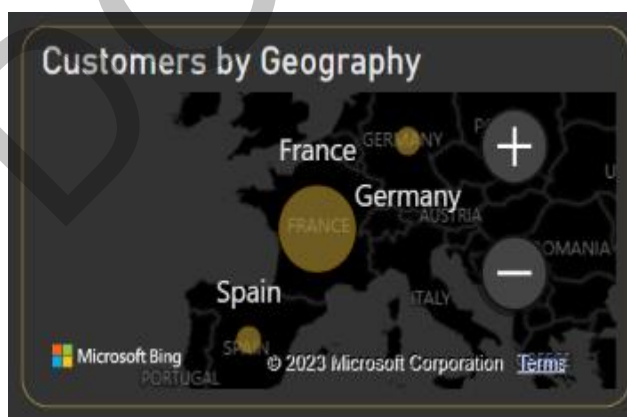


Figure 3: Customer by Geography

Title	Customer by Geography
Chart Type	Map
Reason	To give the distribution of customer base (quantity) across countries in the map (categorical) using size of the bubbles in comparison
Findings	Half of the customers are resident in France, while Germany and Spain have equal distribution of overall

b. Customer Engagements: credit card status, active status, churn status, tenure

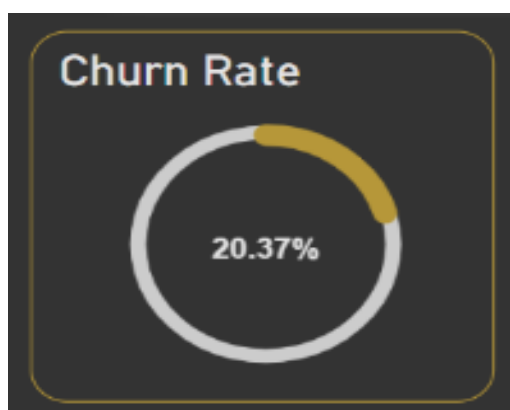


Figure 4: Customer by Churn Status

Title	Churn Rate
Chart Type	KPI Donut Chart
Reason	To give the rate or level of those that left vs those that remained by percentage of exited customers and retained
Findings	Majority (80%) of the customers are still with the bank, while 20% (2,037) of the overall customer left the bank

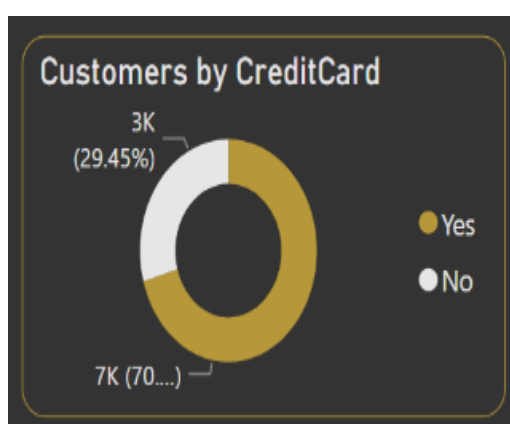


Figure 5: Customer by Credit Card Status

Title	Customer by CreditCard
Chart Type	Donut Chart
Reason	To give the spread of customer based on their credit card status (i.e., those that have credit card vs those that don't) by number and percentage of each category
Findings	Approximately 7 in 10 customers have credit card across the customer base, while 30% are not subscribing to it

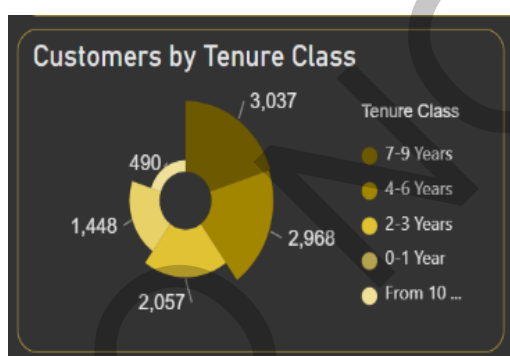


Figure 6: Customer by Tenure Class

Title	Customer by Tenure Class
Chart Type	Aster Plot Chart
Reason	To give the distribution of customer base (quantity) across different tenure class (categorical) using sizes of the blades in comparison
Findings	Approximately 60% of customers have stayed more than 3 years since they joined the bank

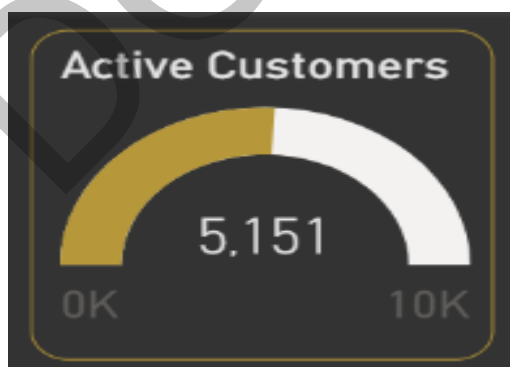


Figure 7: Gauge of Active Customers

Title	Active Customers
Chart Type	Gauge Axis Chart
Reason	To give the rate, level, or proportion of those that are active against the overall customer base. In cases where there are external indices to compare with, this could be one way to factor those significant differences
Findings	Slightly more than 50% of customers are active

c. Financial Status: tone of deposits (balance), income (salary), credit score



To get the answers to this question, measures were created and put on card. They also formed the KPIs for other level of analytics in the data story.

Total Balance = sum(Churn_Modelling[Balance])

→ variable name = Total Balance; function = sum; Table name = Churn_Modelling; Column name = Balance. Similarly,

*Average Salary = AVERAGE(Churn_Modelling[EstimatedSalary]) and
Avg. CreditScore = AVERAGE(Churn_Modelling[CreditScore])*

Total Balance was calculated to get the overall net deposits (balance) – which is one of the metrics to underscore the performances of the bank generally, and specifically use it to measure the impacts of those that have left. Same way for Average Salary and Credit Score, but the former would help to understand the feasible income and future balance of the customers in cases where there is direct correlation between salary and balance; while the later will give the best attribute for marketing other bank products like car finance, credit card, mortgages, thereby building the bank's portfolio.

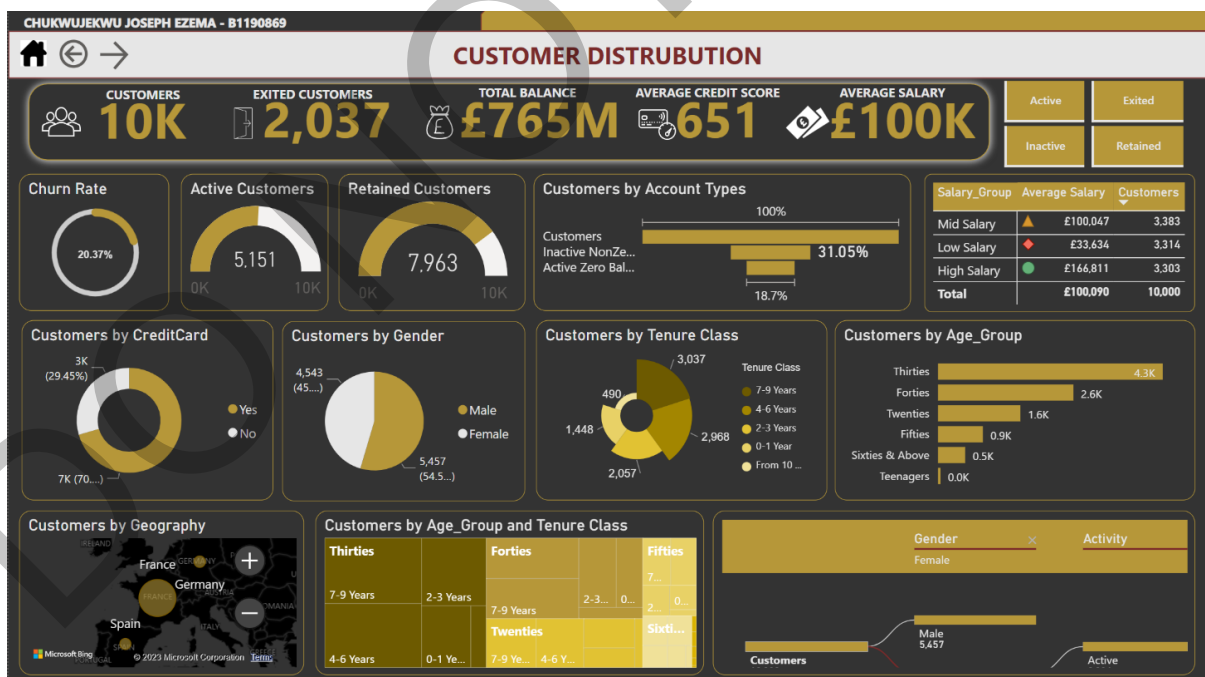


Figure 8: Full Dashboard for Customer Distribution

In general, this dashboard has gone extra to give other dimensions of the customer base like Customer by Account Types, distribution of customer across salary ranges, and with filter to dive deep into the analytics for the end users to have full understanding of the analytics.

Q2 Is there any significance of those that left vs those that remained when compared:

This second question takes us into the specifics of this whole report – which is practically a comparative analysis of customers that left versus those that are still with the bank to understand the impact of the churn rate.

a. Churn Distribution across age group, gender, geography

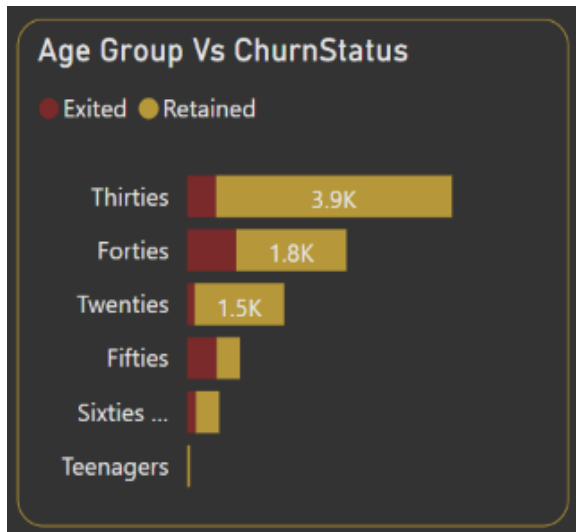


Figure 9: Churn Distribution across Age Groups

Title	Age_Group Vs ChurnStatus
Chart Type	Horizontal Stacked Bar Chart
Reason	To give the distribution of customer (quantity) in terms of their churn status (legend) across different age groups (categorical) using lengths and colours of the bars in comparison
Findings	Out of 4.3k customers in their thirties, 3.9k are still with the bank while Forties is the age group where most of churn rate is coming from at 806

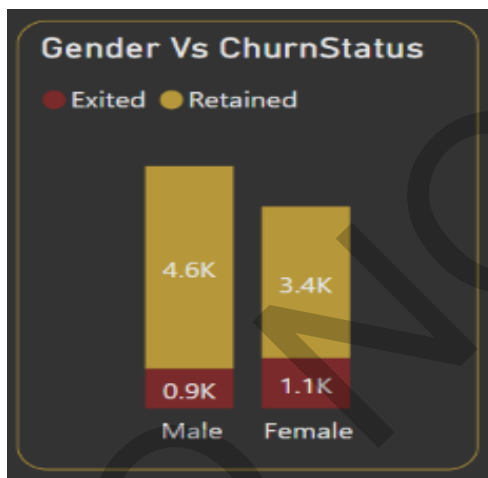


Figure 10: Churn Distribution across Gender

Title	Gender Vs ChurnStatus
Chart Type	Vertical Stacked Bar Chart
Reason	To give the distribution of customer (quantity) in terms of their churn status (legend) across gender (categorical) using heights and colours of the bars in comparison
Findings	Out of 5.4k male customers in the bank, 4.6k are still with the bank while the females are the most amongst those that left with about 200 more customer

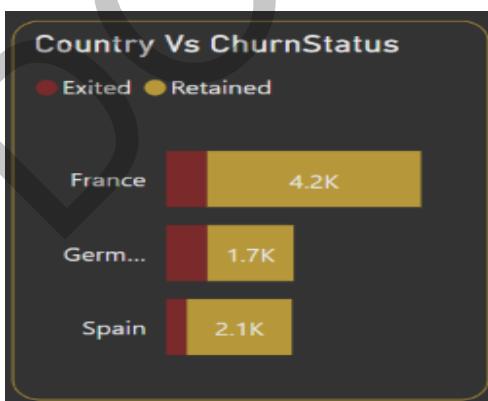


Figure 11: Churn Distribution across Country

Title	Country Vs ChurnStatus
Chart Type	Horizontal Stacked Bar Chart
Reason	To give the distribution of customer (quantity) in terms of their churn status (legend) across country (categorical) using lengths and colours of the bars in comparison
Findings	France has the most base but Germany has 814 leavers against 810 from France

b. Churn Distribution across engagements: credit card, activity, accounts, tenure

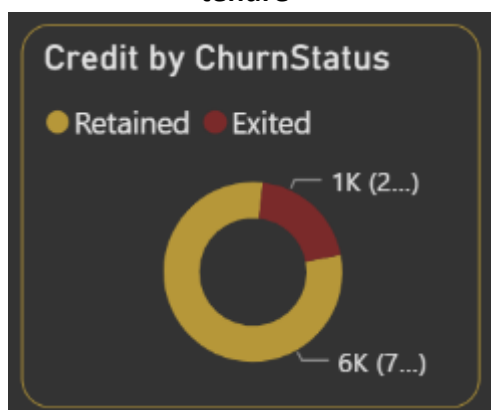


Figure 12: Churn Distribution across Customers with Credit Card

Title	Credit by ChurnStatus
Chart Type	Donut Chart
Reason	To give the distribution of customers with credit card(quantity) in terms of their churn status (legend) using arcs and colours - given by numbers and percentages of the quantity (measure)
Findings	Out of 7k customers with credit card, 1k of them left while 6k are still with bank

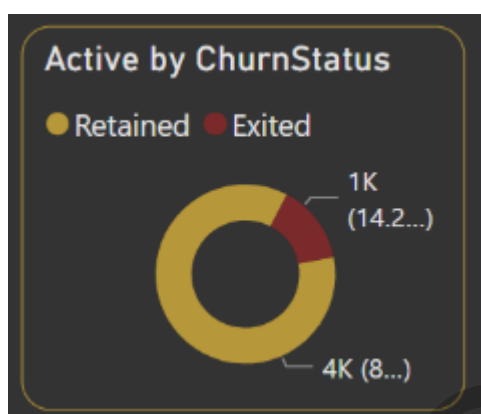


Figure 13: Churn Distribution across Active Customers

Title	Active by ChurnStatus
Chart Type	Donut Chart
Reason	To give the distribution of active customers (quantity) in terms of their churn status (legend) using arcs and colours - given by numbers and percentages of the quantity (measure)
Findings	Out of 5k active customers, 1k of them left while 4k are still with bank

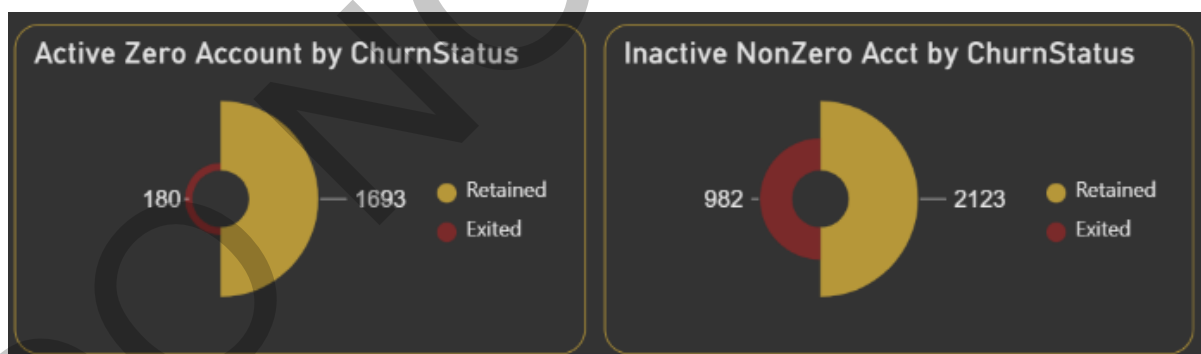


Figure 14: Churn Distribution across Active Zero and Inactive NonZero Account Types

Title	Active Zero vs Inactive NonZero Accounts by ChurnStatus
Chart Type	Aster Plot Chart
Reason	To give the distribution of active zero account customers (left measure) and inactive non-zero account customers (right measure) in terms of their churn status (legend) using arcs and colours - given by numbers and percentages of the quantity (measure)
Findings	From 1,873 active zero accounts, 180 of them belong to those that left the bank, while out of 3,105 inactive non-zero accounts, 982 are have exited

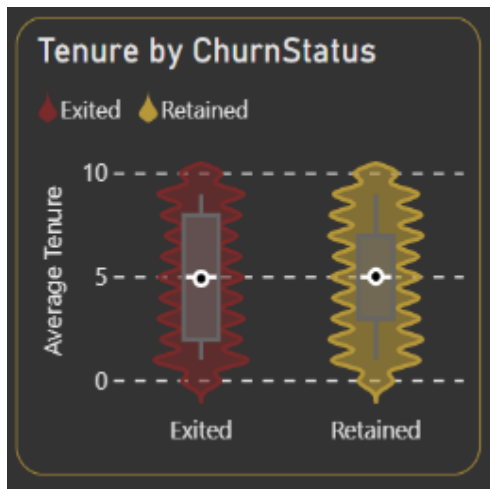


Figure 15: Average Tenure across the Churn Status

Title	Average Tenure by ChurnStatus
Chart Type	Violin Plot Chart
Reason	To give the numerical distribution of average tenure (quantity) across the churn status (categories) using mean box – traceable on the y-axis. This chart is fit to give statistical summary of a numerical distribution.
Findings	The concentration of customers that left have lasted between 2 to 8 years with the bank while the Retained customers are 3 to 7 years. But at 5 years average

c. Churn comparisons across the financial status: balance, salary, credit score

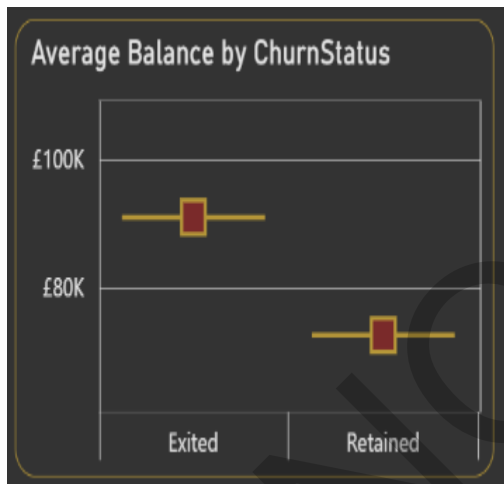


Figure 16: Average Balance by Churn Status

Title	Average Balance by ChurnStatus
Chart Type	Box and Whisker Plot Chart
Reason	To give the numerical distribution of average balance (quantity) across the churn status (categories) using mean 'square' – traceable on the y-axis. This chart is another chart fit to give statistical summary of a numerical distribution.
Findings	The average balance across the churn status shows that though that number of those that left the bank is 2k (20%), majority of them had very good balance because 'Exited' has 20k more balance

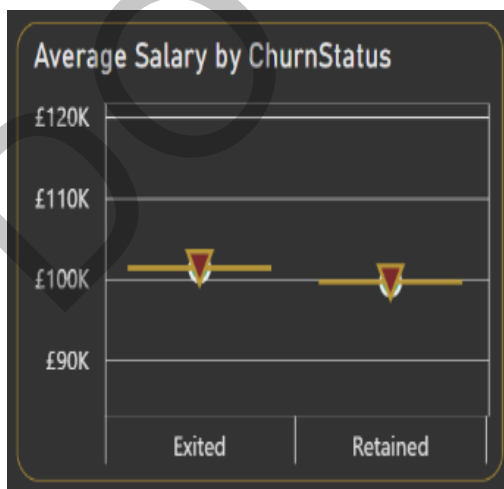


Figure 17: Average Salary by Churn Status

Title	Average Salary by ChurnStatus
Chart Type	Box and Whisker Plot Chart
Reason	To give the numerical distribution of average salary (quantity) across the churn status (categories) using mean 'triangle' – traceable on the y-axis. This chart is another chart fit to give statistical summary of a numerical distribution.
Findings	There is no significance in salaries across since everybody is at nearly £100k

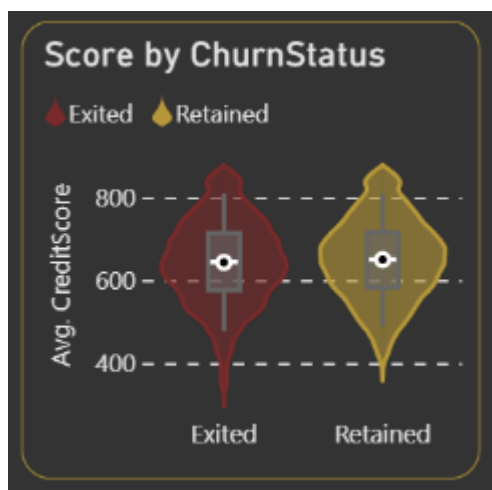


Figure 18: Average Credit Score by Churn Status

Title	Average Credit Score by ChurnStatus
Chart Type	Violin Plot Chart
Reason	To give the numerical distribution of average credit score (quantity) across the churn status (categories) using mean whisker box – traceable on the y-axis. This chart was also used to understand the statistical summary of the numerical distribution
Findings	There is no big difference in credit score across since everybody is at 645 average

During this analysis, measures were created to further analyse some the categories in comparative analysis. Measures like Active, Credit, Averages of the numerical columns and so on – which will be discussed in full in the DAX section of the Appendix.

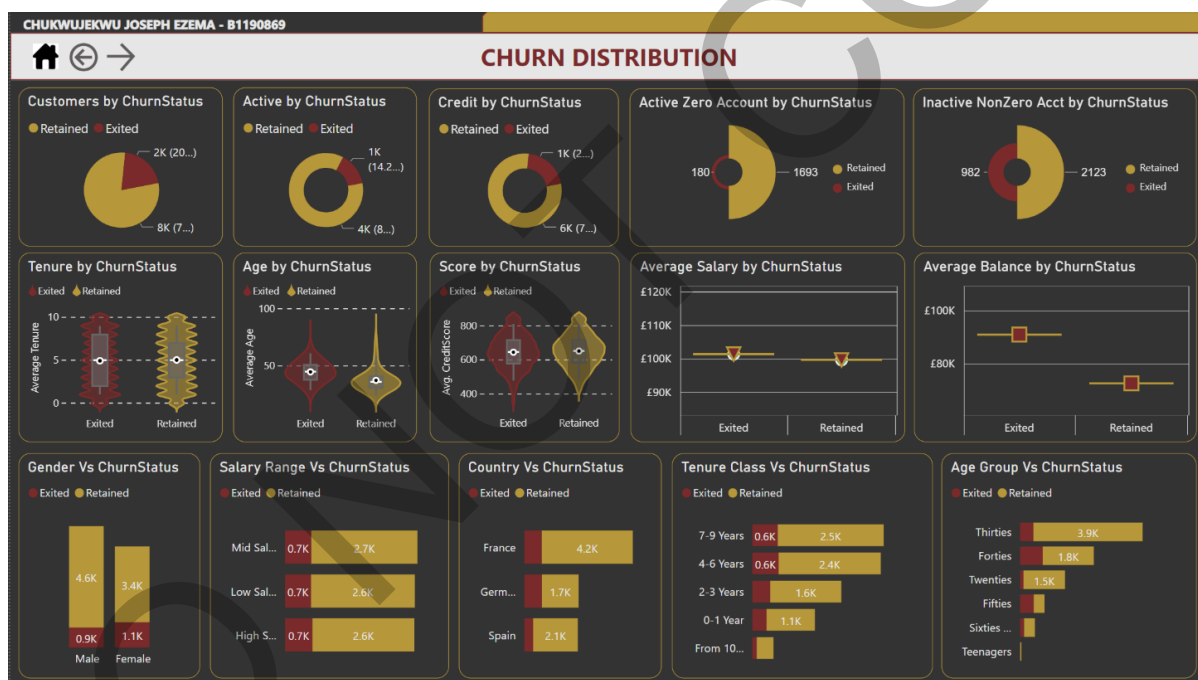


Figure 19: Full Dashboard of the Churn Distribution

In general, the problems addressed in Q2 were collectively captured in the second dashboard (figure 19) and third dashboard (figure 20) which give the wider knowledge of the comparative analysis of those that left the bank and customers that are still with the bank.

The clear significance across the dashboard between the 'Exited' and 'Retained' is borne from the root difference that the Exited are 20% of the customer base against 80% of the 'Retained'

In addition, there was an intention to explore some of the advanced charts like Violin Plot and Whisker Box; this was because they are fit for numerical distribution which can then be spread across a target class (ChurnStatus) for optimal comparisons.

d. How does the financial engagements cut across other dimensions: age group, tenure class

We provoked some questions from the last analysis; one of them being how the financial engagements cut across other dimensions: age group, tenure class – regarding the Churn Status.

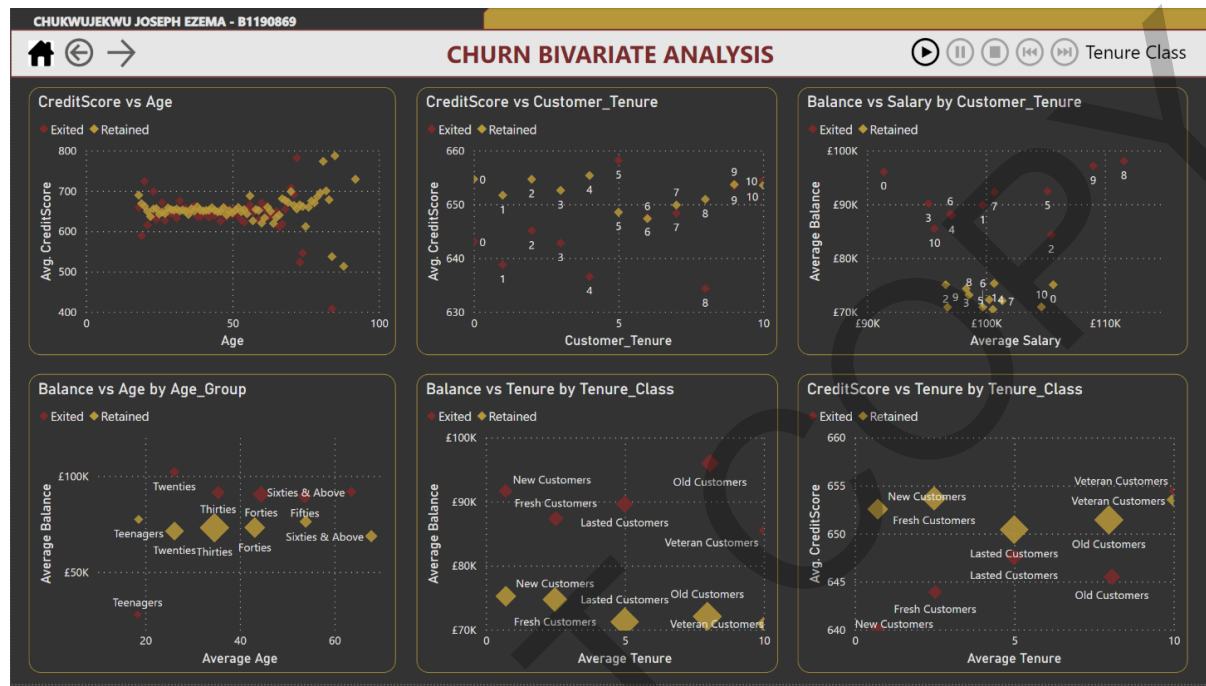


Figure 20: Full Dashboard of Churn Bivariate Analysis

Here, the analysis involves two variables and one or two categorical variable which required a scatterplot chart to visualize it. The focus point (Churn Status) was used to separate the two major types of customers – Exited and Retained, so to compare.

Findings:

- ❖ There is no clear disparity in the ages of customer in 'CreditScore vs Age'. Although fewer older customers that exited had average credit score less than 600
- ❖ In 'CreditScore vs Customer_Tenure', the exited customers with tenure not up to 5 years, considering the gap in between them (difference of 10), had no huge significance across the ChurnStatus, though those that remained had better score.
- ❖ In 'Balance vs Salary by Customer_Tenure', there is clear significance that those that have left the bank have more balance individually and when rated according to tenure
- ❖ Same similar difference applies to 'Balance vs Age by Age_Group' and Balance vs Tenure by Tenure_Class', average balance of exited customers is more than those that remained when rated according to age group and tenure class
- ❖ Just like the point 2, the retained customers had more credit scores but not with a big difference

Q3 What are contributing to the balance of those that left vs those that remained

In this third question, we have used the principle of regression and correlation to analyse factors influencing balance. The visuals selected were to show the mathematical differences.

a. Does the balance of those that left depend on their estimated salaries

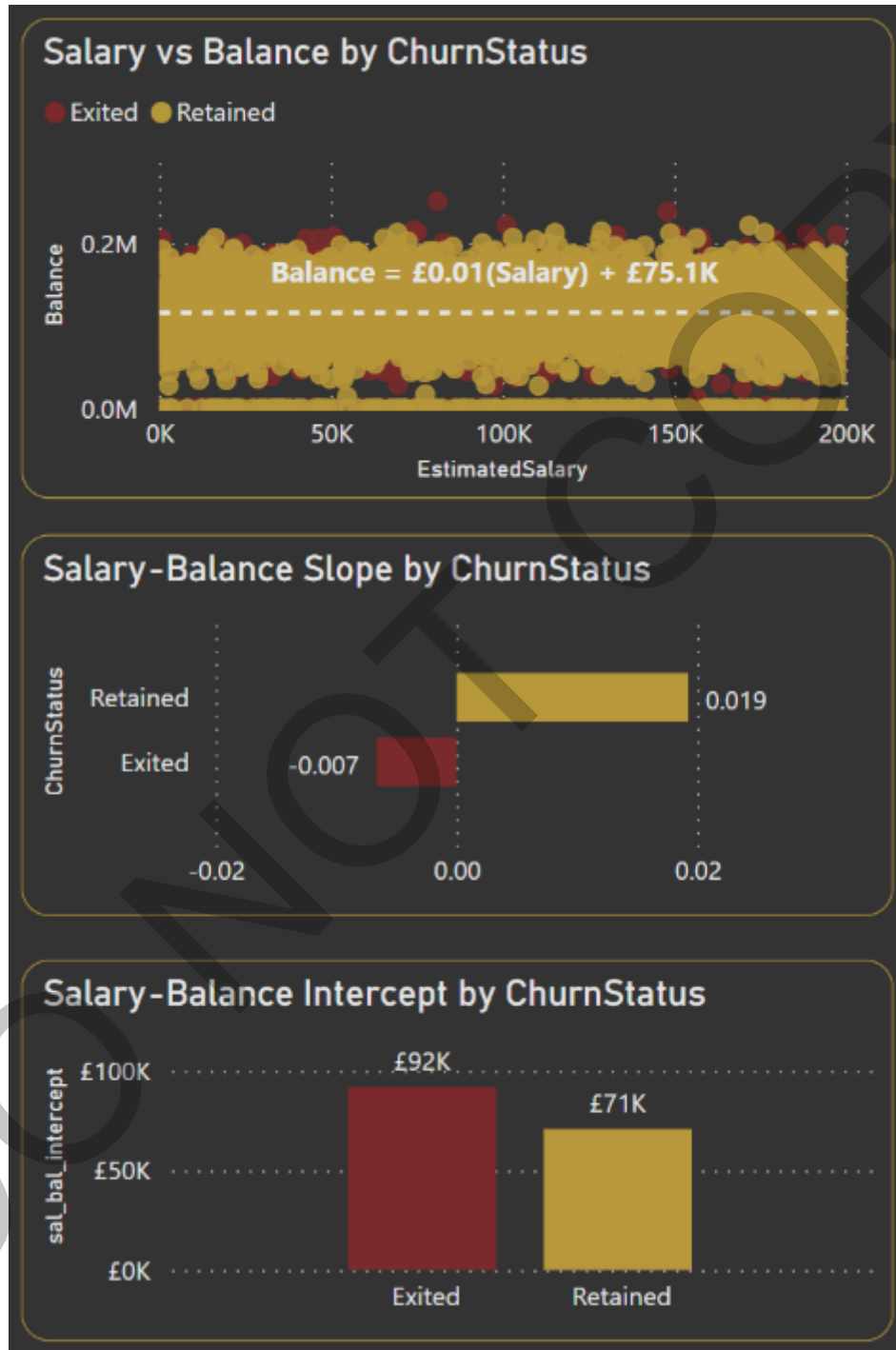


Figure 21: Correlation of Balance to Salary

- ❖ There is no significant correlation between the salary and the balance of customers
- ❖ Those that left the bank have negative correlation of balance to salary at -0.007
- ❖ Those that remained have positive correlation of balance to salary at 0.019

b. How does age affect the balance of those that left the bank

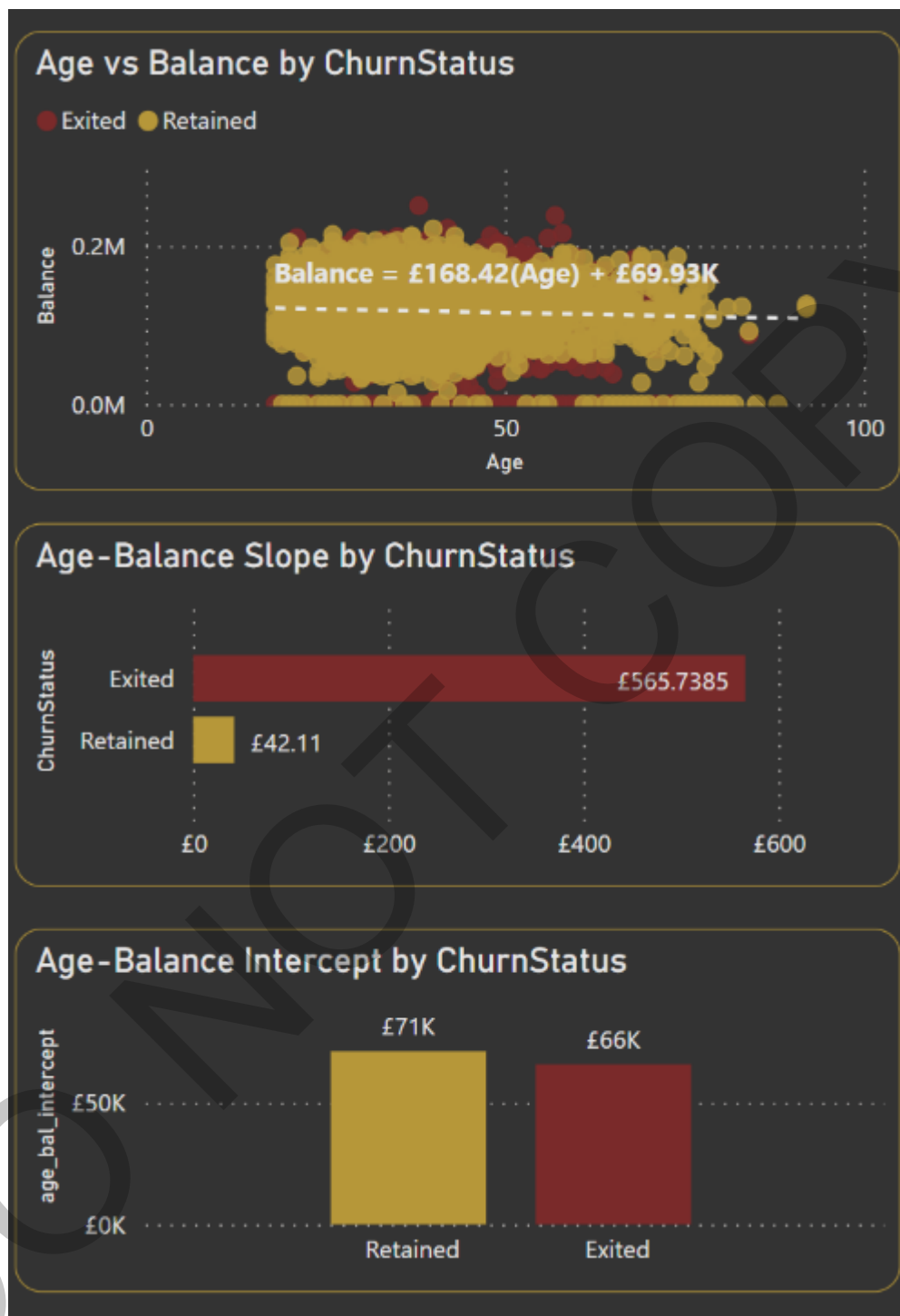


Figure 22: Correlation of Balance to Age

- ❖ There is a significant correlation between the age and the balance of customers at £168.42 for every increase in age. This shows the more the older people patronize them the more there is chances of harbouring large account balance.
- ❖ Those that left the bank have higher correlation of balance to age at £565.74
- ❖ Those that remained have very low correlation of balance to age at £42.11
- ❖ So, there is a huge impact of churn rate considering balance to age correlation

c. Is there a direct correlation between tenures of those that left and their balance

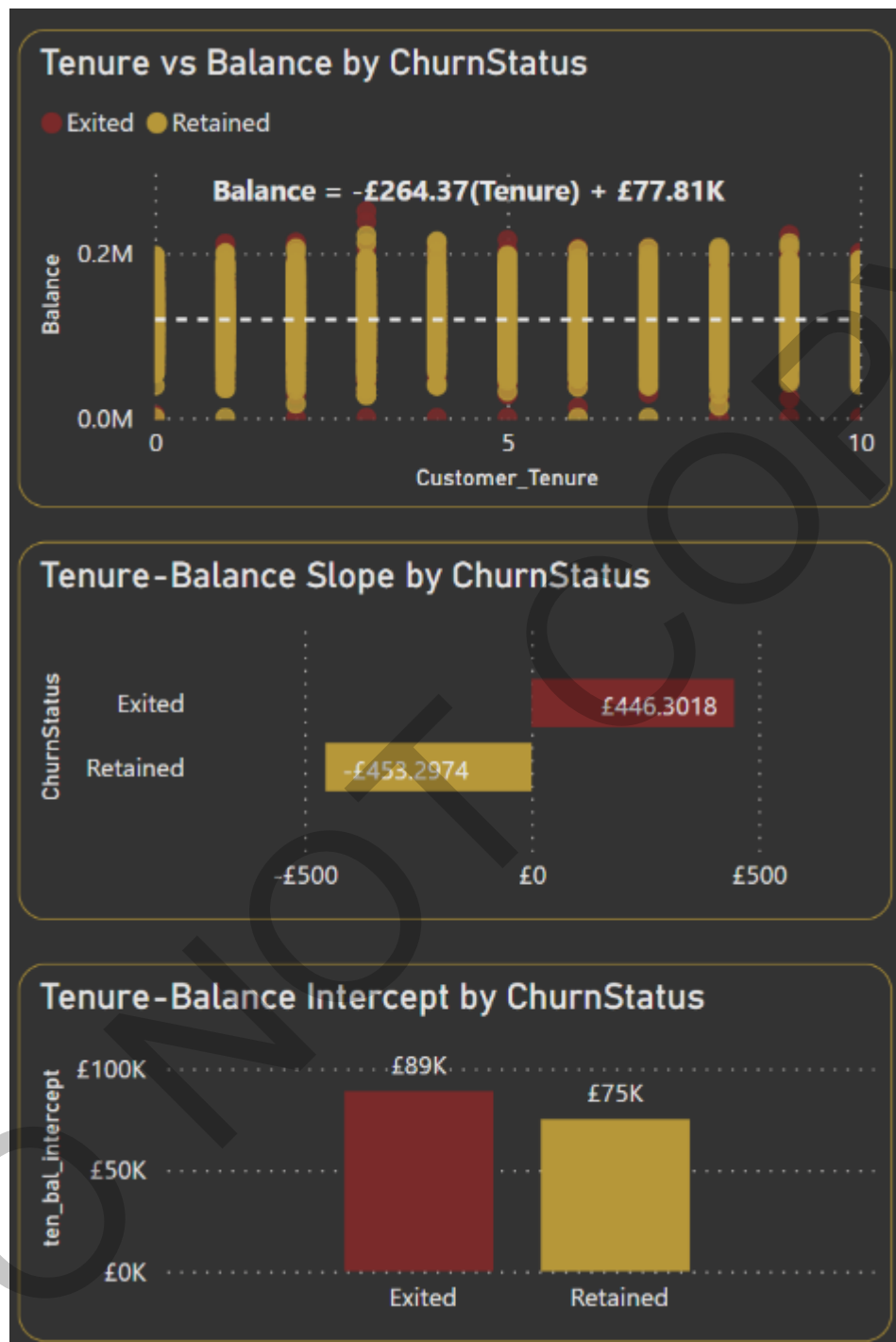


Figure 23: Correlation of Balance to Tenure

- ❖ Contrary to the previous findings, there is a negative correlation between the tenure and the balance of customers at $-\text{£}264.37$ for every increase in tenure. This shows the more people stay and grow with them the more there is reduction in their account balance.
- ❖ However, this is not the case for those that left the bank because they have higher positive correlation of balance to tenure at $\text{£}446.30$ – meaning they keep substantial money in the bank since they joined.

- ❖ Those that remained have the opposite correlation of balance to tenure at -£453.30
- ❖ So, there is a huge impact of churn rate considering balance to tenure correlation

Q4 What are influencing the credit scores of those that left vs those that remained

In the same manner with the analysis of Q3, this seeks to identify with the factors affecting the credit scores.

a. How does balance affect the credit score of those that left the bank

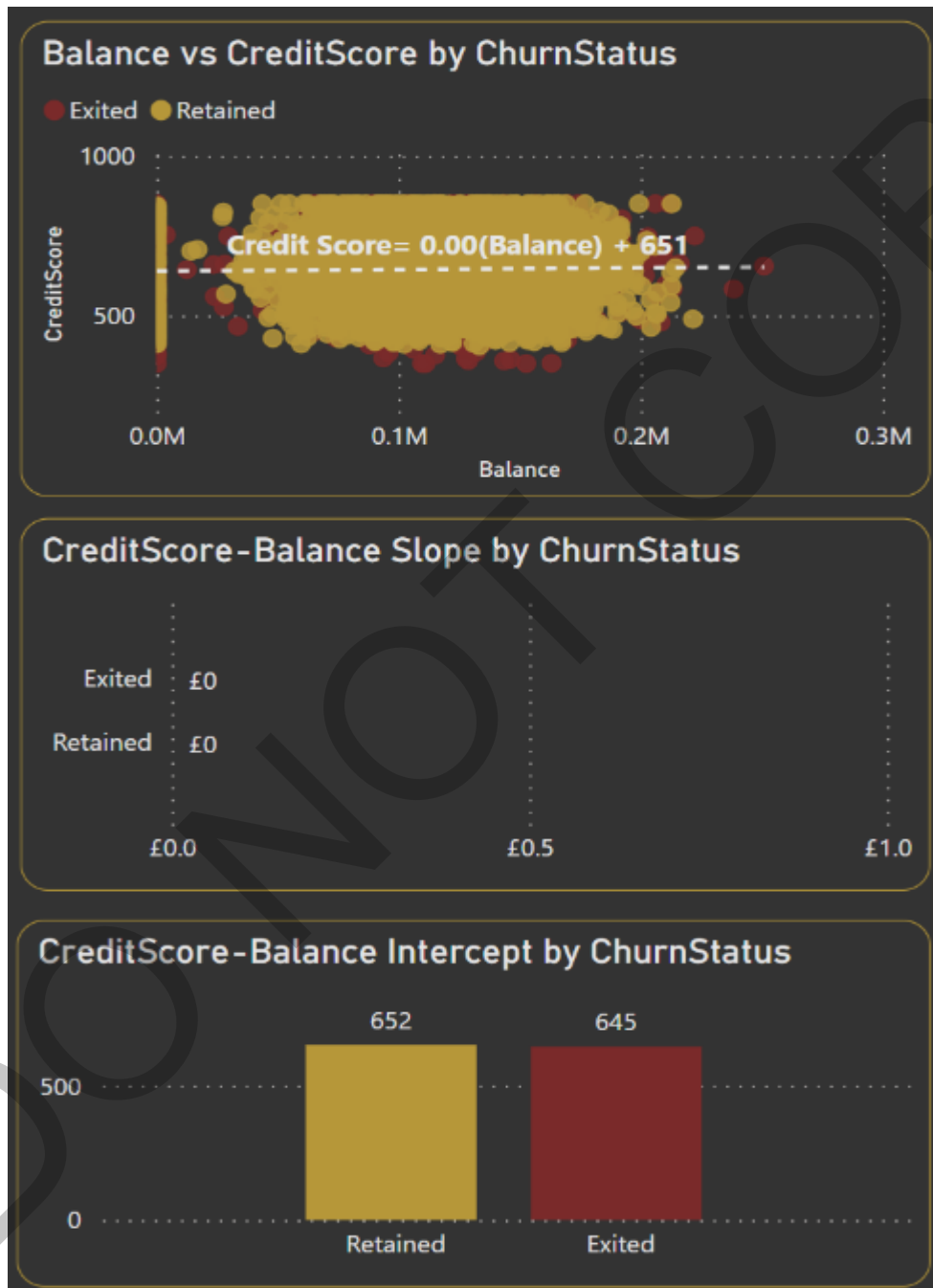


Figure 24: Correlation of Credit Score to Balance

- ❖ There is no correlation between the credit score and the balance of customers
- ❖ The Exited vs Retained showed no correlation of credit score to balance
- ❖ So, Balance is not a good measure for determining credit scores of customers

b. Is there a direct correlation between tenures of those that left and their credit scores

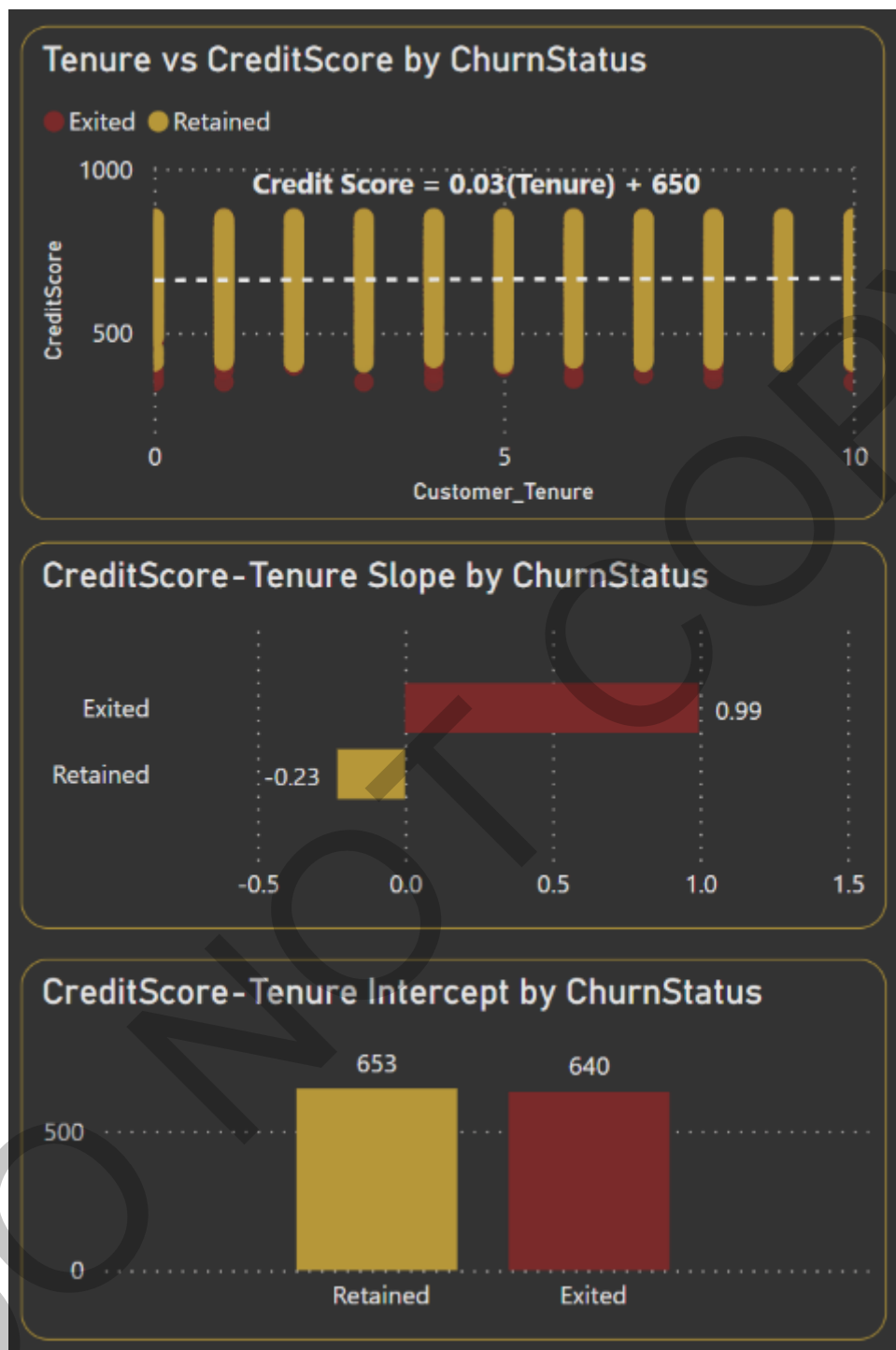


Figure 25: Correlation of Credit Score to Tenure

- ❖ There is no significant correlation between the credit score and tenure of customers
- ❖ Although the Exited Correlation showed that there is approximately increase in the credit scores by 1 as tenure increases.
- ❖ While the Retained Correlation showed negative correlation of credit score to tenure
- ❖ On average, the credit scores of both classes are at barest minimum of 640 (intercept)
- ❖ So, Tenure is not good enough to measure the determining factors of credit scores

c. Does the credit scores of those that left depend on their estimated salaries

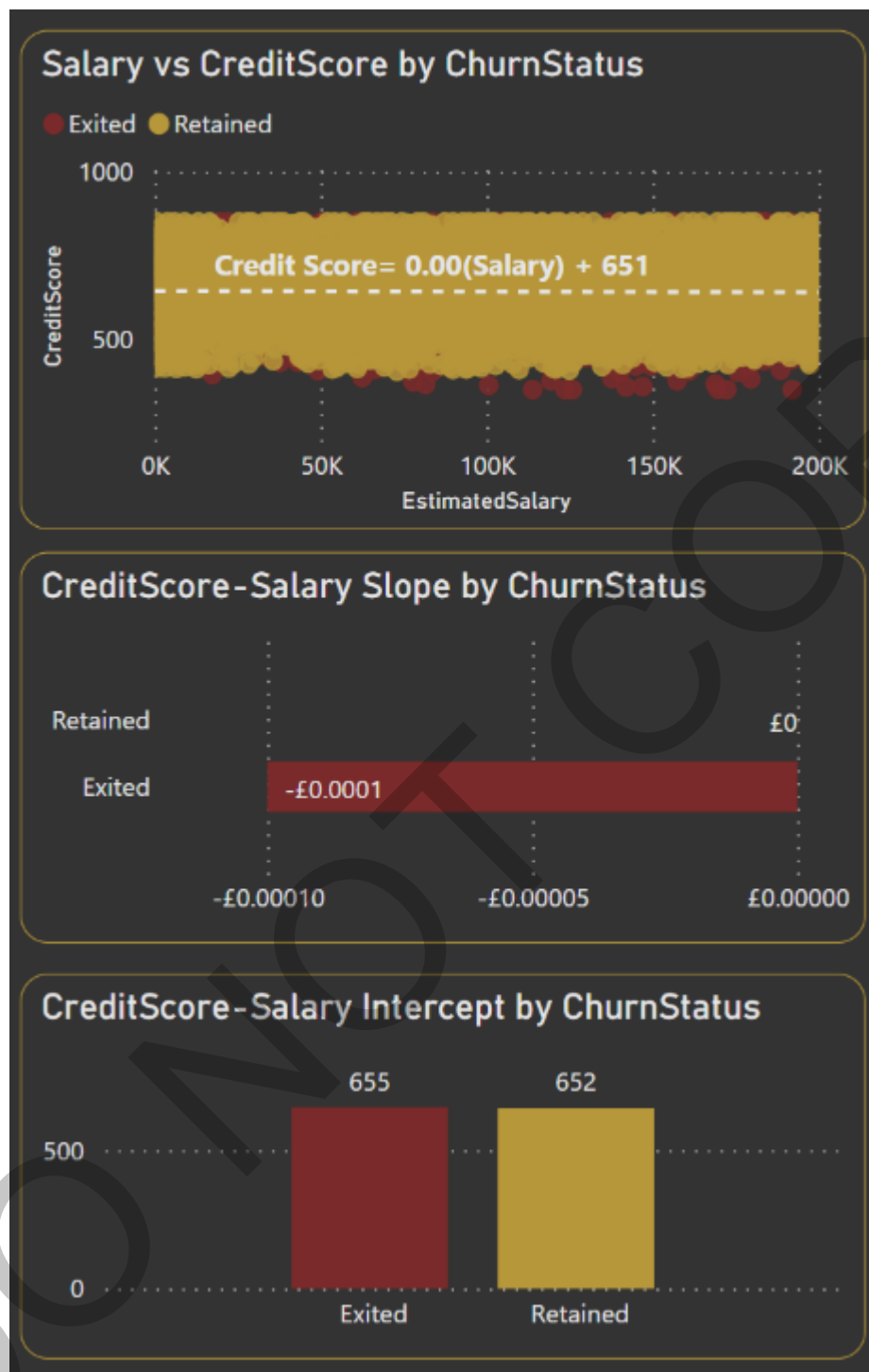


Figure 26: Correlation of Credit Score to Salary

- ❖ There is no correlation between the credit score and the salary of customers
- ❖ The Exited vs Retained showed no correlation of credit score to salary in a justifiable sense
- ❖ So, Salary is not a good measure for determining credit scores of customers
- ❖ Although on average, the credit scores of both classes are at barest minimum of 652 (intercept)

Q5 Considering some key factors, what is the impact of customer churn rate:

In the actual sense, this is the most important question this report seeks to answer. And that is to understand the impact of those that left the in some of the key areas like customer base, total account portfolio of the bank in form of total balance, activities, and marketing plans.

a. Does it affect the customer base

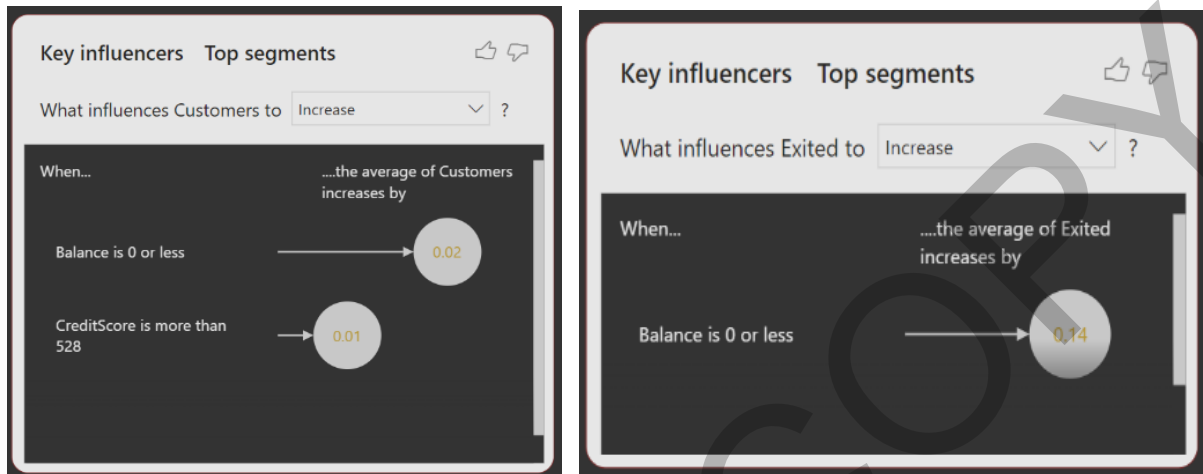


Figure 27: AI Insights on Factors affecting Customer Base

- ❖ Using AI-enabled charts (Key Influencer Chart), there is a common pendulum tied to overall customer base and exited customers; and that is when balance tends to be low. In other words, if what affects the customer base to increase is same thing that affects the number of exited customers to increase, the more the bank grows their customer base, the more customers continue to leave the bank. This is a big impact until the correlation is cut off

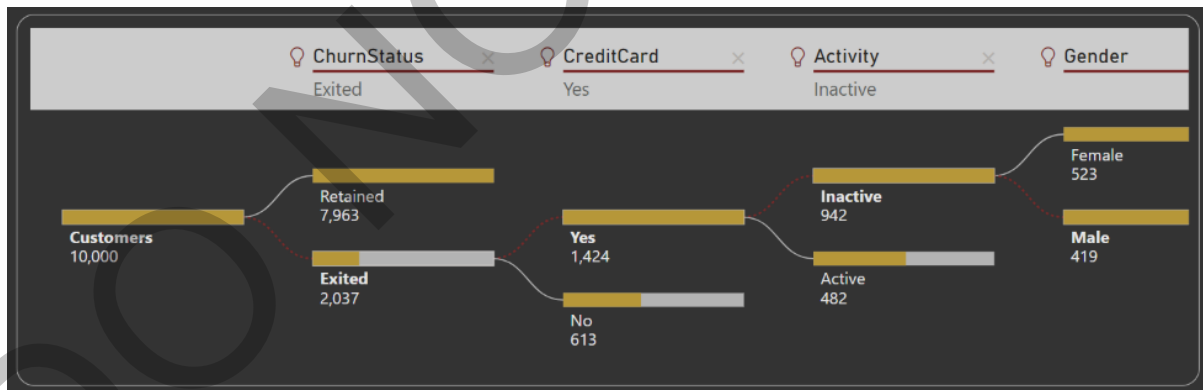


Figure 28: AI-enabled Decomposition of the Customer Base

- ❖ In relative comparisons of the churn status, credit card, activity, and gender – to the customer base of the bank, churn status is considered to have most importance. This means that churn rate, whether high or low, has a way of affecting the overall customer base
- ❖ Majority of those that left have credit card – which is more than twice of those that don't
- ❖ Therefore, considering these factors, it's safe to say the exited customers, though at 20% rate, had impact in the customer base

b. Does it affect the current balance in the bank

From the findings of Q3, the analysis showed that the exited customers have more account balance that those still with the bank when rated according to the tenure class and age group.

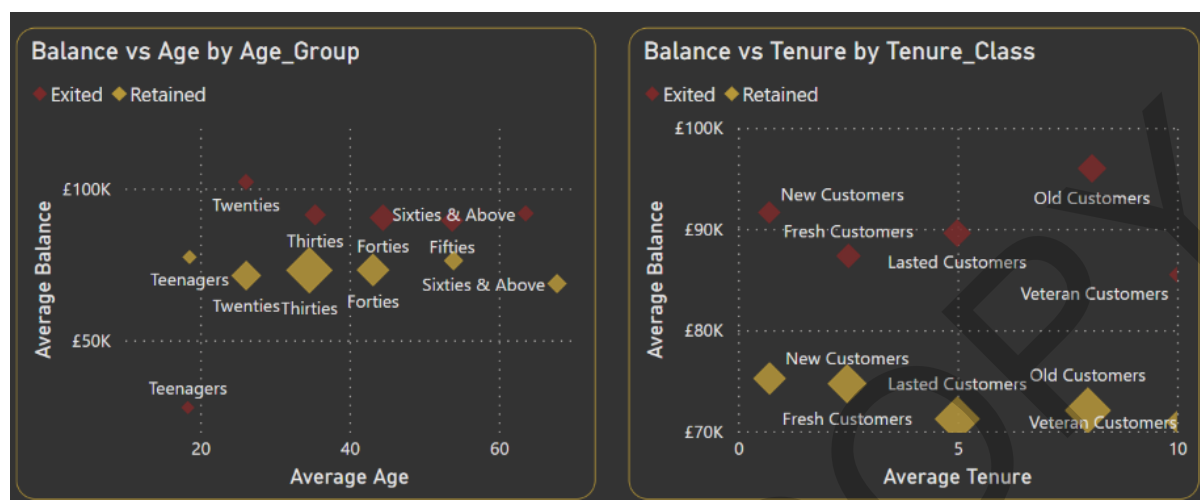


Figure 29: Charts showing how the current balance is affected by the Churn Rate

- ❖ By all age groups, except the Teenagers, the average balance of exited customers was higher than those that remained. This is a huge impact on the portfolio of the bank
- ❖ By all tenure class, none of the class that left the bank had an average balance below £80k while it's the opposite for the Retained because none of their tenure class passed an average balance of same £80k.

c. Does it reduce the activeness of customers



Figure 30: AI-enabled KPI showing how Churn Rate does not reduce the activeness of customers

- ❖ What influences Active customers to increase is when their credit score is more than 849 or they come from France
- ❖ which is opposite of majority of customers that exited the bank.
- ❖ In addition, in figure 28, there are more inactive accounts of customers that exited than those that remained.
- ❖ Therefore, the churn rate doesn't affect activeness of customers in the bank

D. CONCLUSIONS AND RECOMMENDATIONS

Summary

- ❖ There are 10,000 customers as we have seen from the data source
- ❖ More Males than Females according to the Figure 1, with about 10% gap
- ❖ Almost half of customers is in their thirties, followed by the Forties slightly above 25% overall. Teenagers are the least age group
- ❖ Majority (80%) of the customers are still with the bank, while 20% (2,037) of the overall customer left the bank
- ❖ Approximately 7 in 10 customers have credit card across the customer base, while 30% are not subscribing to it
- ❖ Approximately 60% of customers have stayed more than 3 years since they joined the bank
- ❖ Out of 4.3k customers in their thirties, 3.9k are still with the bank while Forties is the age group where most of churn rate is coming from at 806 exited customers
- ❖ France has the highest base, but Germany has 814 leavers against 810 from France
- ❖ Out of 7k customers with credit card, 1k of them left while 6k are still with bank
- ❖ From 1,873 active zero accounts, 180 of them belong to those that left the bank, while out of 3,105 inactive non-zero accounts, 982 are for the leavers
- ❖ There is no significant correlation between the salary and the balance of customers
- ❖ There is a significant correlation between the age and the balance of customers at £168.42 for every increase in age. This shows the more the older people patronize them the more there is chances of harbouring large account balance.
- ❖ Those that left the bank have higher correlation of balance to age at £565.74
- ❖ Contrary to the previous findings, there is a negative correlation between the tenure and the balance of customers at -£264.37 for every increase in tenure. This shows the more people stay and grow with them the more there is reduction in their account balance.
- ❖ However, this is not the case for those that left the bank because they have higher positive correlation of balance to tenure at £446.30 – meaning they keep substantial money in the bank since they joined
- ❖ There is no correlation between the credit score and the balance of customers
- ❖ The Exited vs Retained showed no correlation of credit score to balance
- ❖ There is no significant correlation between the credit score and tenure of customers
- ❖ Although the Exited Correlation showed that there is approximately increase in the credit scores by 1 as tenure increases
- ❖ Salary is not a good measure for determining credit scores of customers
- ❖ In relative comparisons of the churn status, credit card, activity, and gender – to the customer base of the bank, churn status is considered to have most importance. This means that churn rate, whether high or low, has a way of affecting the overall customer base
- ❖ Majority of those that left have credit card – which is more than twice of those that don't

- ❖ By all tenure class, none of the class that left the bank had an average balance below £80k while it's the opposite for the Retained because none of their tenure class passed an average balance of same £80k
- ❖ In addition, in figure 28, there are more inactive accounts of customers that exited than those that remained.
- ❖ The churn rate doesn't affect activeness of customers in the bank
- ❖ Overall, considering these factors, it's safe to say the exited customers, though at 20% rate, had impact in the customer base

Recommendations

- ❖ Since there is no impact of the churn rate on the activeness of the customers, the management of the bank should not ensure to convert their inactive accounts to active. Perhaps, those that exited were because of the inactiveness of their accounts
- ❖ The management should build marketing campaigns against zero account balance to their existing customers because there is finding that the tendency of customer to leave increase when their account balance hits zero
- ❖ As there is a positive correlation between age and the account balance, which means older people can get fat figures into the bank, prospective older customers should be targeted
- ❖ It is okay to launch a zero-account opener for new customers since that also increases the customer base. However, strategies should be put in place to grow the accounts
- ❖ If the majority of those that left have credit card – which is more than twice of those that don't, then perhaps the value of the credit card should be revisited or promoted

Personal Conclusion

Before this project, my kind of reports and analysis used to be with basic visuals from the visualization pane in power bi desktop, but this project has exposed me to a lot of advanced usage. Again, I have never bothered to model my data from flat tables into facts and dimensional tables as I have done in this project. It was a big task for me to master at first, but with constant practice I was able to scale through by understanding the concept of diving the queries, getting the ideology of the relationships between tables, and mapping them with keys (table IDs) well in the model view.

One of the key practices I will hold onto from this project is that the relationships amongst tables work better when there are primary keys that connect to each (like an electrical connection). This is what really helped me to grow the confidence needed to work on this project. Another thing is the exploration of the advanced visuals by understanding their usages and maximizing their beauty in the most effective way possible.

Lastly but not the least, is to keep on practising with millions of datasets out there from Kaggle, UCL repository, or other sources, to be a full-fledged Power BI and Big Data Analyst.

BIG DATA AND BUSINESS INTELLIGENCE



Appendix: BI Design

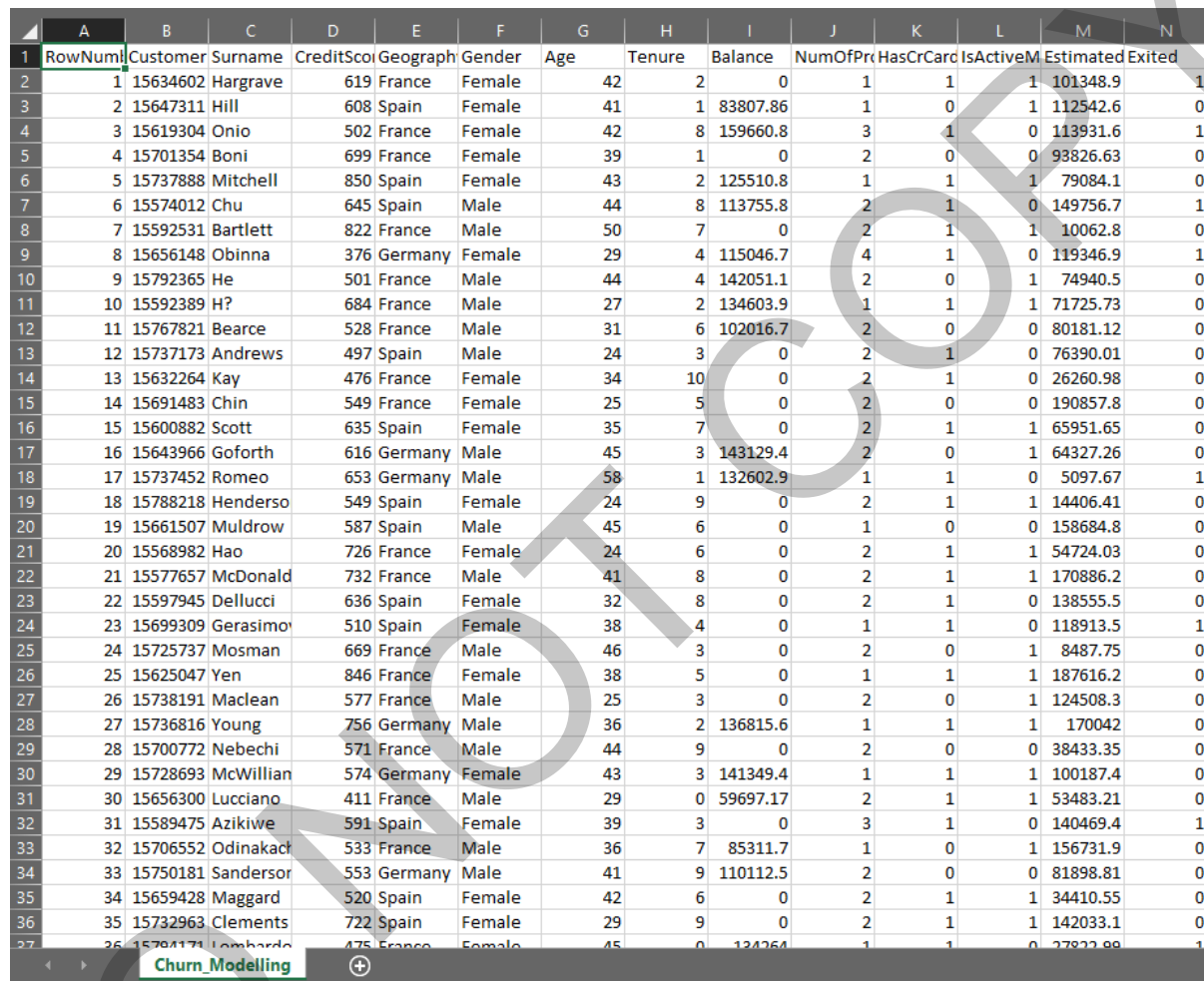


APPENDIX: BI DESIGN

I. DATA PRE-PROCESSING AND DATA CLEANING

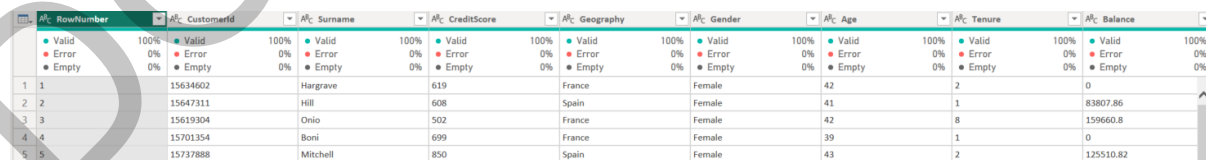
Load and Clean Dataset:

- Loaded the original dataset in csv (after downloading it from the data source as provided in the report) into the power query editor for further exploration and cleaning – using the ‘Get Data’ dropdown with the ‘Text/CSV’ option



RowNumber	CustomerId	Surname	CreditScore	Geography	Gender	Age	Tenure	Balance	NumOfProducts	HasCreditCard	IsActiveMember	EstimatedSalary	Exited
1	15634602	Hargrave	619	France	Female	42	2	0	1	1	1	101348.9	1
2	15647311	Hill	608	Spain	Female	41	1	83807.86	1	0	1	112542.6	0
3	15619304	Onio	502	France	Female	42	8	159660.8	3	1	0	113931.6	1
4	15701354	Boni	699	France	Female	39	1	0	2	0	0	93826.63	0
5	15737888	Mitchell	850	Spain	Female	43	2	125510.8	1	1	1	79084.1	0
6	15574012	Chu	645	Spain	Male	44	8	113755.8	2	1	0	149756.7	1
7	15592531	Bartlett	822	France	Male	50	7	0	2	1	1	10062.8	0
8	15656148	Obinna	376	Germany	Female	29	4	115046.7	4	1	0	119346.9	1
9	15792365	He	501	France	Male	44	4	142051.1	2	0	1	74940.5	0
10	15592389	H?	684	France	Male	27	2	134603.9	1	1	1	71725.73	0
11	15767821	Bearce	528	France	Male	31	6	102016.7	2	0	0	80181.12	0
12	15737173	Andrews	497	Spain	Male	24	3	0	2	1	0	76390.01	0
13	15632264	Kay	476	France	Female	34	10	0	2	1	0	26260.98	0
14	15691483	Chin	549	France	Female	25	5	0	2	0	0	190857.8	0
15	15600882	Scott	635	Spain	Female	35	7	0	2	1	1	65951.65	0
16	15643966	Goforth	616	Germany	Male	45	3	143129.4	2	0	1	64327.26	0
17	15737452	Romeo	653	Germany	Male	58	1	132602.9	1	1	0	5097.67	1
18	15788218	Henderso	549	Spain	Female	24	9	0	2	1	1	14406.41	0
19	15661507	Muldrow	587	Spain	Male	45	6	0	1	0	0	158684.8	0
20	15568982	Hao	726	France	Female	24	6	0	2	1	1	54724.03	0
21	15577657	McDonald	732	France	Male	41	8	0	2	1	1	170886.2	0
22	15597945	Dellucci	636	Spain	Female	32	8	0	2	1	0	138555.5	0
23	15699309	Gerasimo	510	Spain	Female	38	4	0	1	1	0	118913.5	1
24	15725737	Mosman	669	France	Male	46	3	0	2	0	1	8487.75	0
25	15625047	Yen	846	France	Female	38	5	0	1	1	1	187616.2	0
26	15738191	Maclean	577	France	Male	25	3	0	2	0	1	124508.3	0
27	15736816	Young	756	Germany	Male	36	2	136815.6	1	1	1	170042	0
28	15700772	Nebechi	571	France	Male	44	9	0	2	0	0	38433.35	0
29	15728693	McWilliam	574	Germany	Female	43	3	141349.4	1	1	1	100187.4	0
30	15656300	Lucciano	411	France	Male	29	0	59697.17	2	1	1	53483.21	0
31	15589475	Azikiwe	591	Spain	Female	39	3	0	3	1	0	140469.4	1
32	15706552	Odinakach	533	France	Male	36	7	85311.7	1	0	1	156731.9	0
33	15750181	Sandersor	553	Germany	Male	41	9	110112.5	2	0	0	81898.81	0
34	15659428	Maggard	520	Spain	Female	42	6	0	2	1	1	34410.55	0
35	15732963	Clements	722	Spain	Female	29	9	0	2	1	1	142033.1	0
36	15704171	Lombardo	475	France	Female	45	0	124264	1	1	0	27822.98	1

Figure 31: Original Dataset (csv) opened from Excel before loading into Power BI



RowNumber	CustomerId	Surname	CreditScore	Geography	Gender	Age	Tenure	Balance	NumOfProducts	HasCreditCard	IsActiveMember	EstimatedSalary	Exited
1	15634602	Hargrave	619	France	Female	42	2	0	1	1	1	101348.9	1
2	15647311	Hill	608	Spain	Female	41	1	83807.86	1	0	1	112542.6	0
3	15619304	Onio	502	France	Female	42	8	159660.8	3	1	0	113931.6	1
4	15701354	Boni	699	France	Female	39	1	0	2	0	0	93826.63	0
5	15737888	Mitchell	850	Spain	Female	43	2	125510.8	1	1	1	79084.1	0

Figure 32: Original Dataset immediately after loading into the Power Query (NB: the view doesn't show the full columns)

- Took note of the number of columns and count of rows: 14 columns by 10,000 rows
- Checked for missing values but there was none; this can be seen from the green lines under the column names indicating all fields are filled or column quality of Valid vs Empty
- Checked for duplicated rows but there was none; all 10k rows were all unique

- Checked if the automated data types matched appropriately (Power BI sometimes promotes the header and converts to appropriate data types for each column). However, I had to check across the columns to be sure.
- Removed unwanted column - which was 'RowNumber' as at this point

1.2 Customerid	1.2 Surname	1.2 CreditScore	1.2 Geography	1.2 Gender	1.2 Age	1.2 Tenure	1.2 Balance
Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%
1	15634602 Hargrave	619	France	Female	42	2	0
2	15647311 Hill	608	Spain	Female	41	1	83807.86
3	15619304 Onlo	502	France	Female	42	8	159660.8
4	15701354 Boni	699	France	Female	39	1	0
5	15737888 Mitchell	850	Spain	Female	43	2	125510.82
6	15574012 Chu	645	Spain	Male	44	8	113755.78
7	15592531 Bartlett	822	France	Male	50	7	0
8	15656148 Obinna	376	Germany	Female	29	4	135046.74
9	15792365 He	501	France	Male	44	4	142051.07
10	15592389 H?	684	France	Male	27	2	134603.88
11	15767821 Bearce	528	France	Male	31	6	102016.72
12	15737173 Andrews	497	Spain	Male	24	3	0
13	15632264 Kay	476	France	Female	34	10	0
14	15691483 Chin	549	France	Female	25	5	0
15	15600882 Scott	635	Spain	Female	35	7	0
16	15643966 Goforth	616	Germany	Male	45	3	143129.41
17	15737452 Romeo	653	Germany	Male	58	1	132602.88
18	15788218 Henderson	549	Spain	Female	24	9	0
19	15661507 Muldrow	587	Spain	Male	45	6	0
20	15568982 Hao	726	France	Female	24	6	0
21	15577657 McDonald	732	France	Male	41	8	0
22	15597945 Dellucci	636	Spain	Female	32	8	0
23	15699309 Gerasimov	510	Spain	Female	38	4	0
24	15725737 Mosman	669	France	Male	46	3	0
25	15625047 Yen	846	France	Female	38	5	0
26	15738191 Maclean	577	France	Male	25	3	0
27	15736816 Young	756	Germany	Male	36	2	136815.64
28	15700772 Nebechi	571	France	Male	44	9	0
29	15728693 McWilliams	574	Germany	Female	43	3	141349.43
30	15656300 Lucciano	411	France	Male	29	0	59697.17
31	15589475 Azikiwe	591	Spain	Female	39	3	0
32	15706552 Odinakachukwu	533	France	Male	36	7	85311.7
33	15750181 Sanderson	533	Germany	Male	41	9	110112.54
34	15659428 Maggard	520	Spain	Female	42	6	0
35	15732963 Clements	722	Spain	Female	29	9	0

Figure 33: Dataset after the first basic data cleaning before duplicating into different queries

Creating Dimensional Tables for Relationships:

- Duplicated the data in Figure 33 into several tables (queries) using the 'Duplicate' option from the right-click menu selection of the data and renamed them according to different dimensions and logical flow of the analytics.

Churn_Modelling
Geography
Gender
IsActive
HasCredit
Churn
Age & Tenure

Figure 34: List of Created Tables originally from the flat table for Data Modelling

For Geography and Gender Tables:

- From the data in Figure 33, removed all other columns except the 'Geography' column by selecting the 'Geography' column, clicking the 'Remove Columns' dropdown button on the 'Home' tab and selecting the 'Remove Other Columns' option

- Removed Duplicates by right-clicking on the column and selecting the 'Remove Duplicates' option
- Added index column from the 'Add Column' tab with first row starting from 11 and increases by 1
- Renamed the index column to GeographyID
- Reordered the GeographyID to be the first column
- Repeated the same process for the Gender table except that the indexing started from 0. In other words, 0 = Female, 1 = Male

1 ² ₃ GeographyID		A ^B _C Geography	
Valid	100%	Valid	100%
Error	0%	Error	0%
Empty	0%	Empty	0%
1	11	France	
2	12	Spain	
3	13	Germany	

1 ² ₃ GenderID		A ^B _C Gender	
Valid	100%	Valid	100%
Error	0%	Error	0%
Empty	0%	Empty	0%
1	0	Female	
2	1	Male	

Figure 35: Final view of the Geography and Gender tables

For IsActive, HasCredit, and Churn Tables:

- Similar things were done with that of Geography, the difference was that there was no addition of index column, instead, conditional column was created based on the 1s and 0s from the respective original binominal columns.
- So, for IsActive table, the original column after removing the duplicates, was renamed to be the ID while the conditional column was used to make the main categorical variable.
- This was done so it can serve the purpose of using the ID as table key in data modelling while keeping the natural 1s and 0s.

1 ² ₃ IsActiveID		A ^B _C Activity	
Valid	100%	Valid	100%
Error	0%	Error	0%
Empty	0%	Empty	0%
1	1	Active	
2	0	Inactive	

1 ² ₃ HasCreditID		A ^B _C CreditCard	
Valid	100%	Valid	100%
Error	0%	Error	0%
Empty	0%	Empty	0%
1	1	Yes	
2	0	No	

1 ² ₃ ChurnID		A ^B _C ChurnStatus	
Valid	100%	Valid	100%
Error	0%	Error	0%
Empty	0%	Empty	0%
1	1	Exited	
2	0	Retained	

Figure 36: Final views of IsActive, HasCredit, and Churn tables after modification

For Age and Tenure Table:

- This was a bit different from others because it contained two numerical columns unlike others that had repeating values. Here, each row is unique and therefore no duplicate was removed to avoid data loss or complication.
- This table doesn't need to be merged since it has a similar key (CustomerId) with fact table already

1 ² 3 CustomerId	1 ² 3 Age	1 ² 3 Customer_Tenure
Valid 100%	Valid 100%	Valid 100%
Error 0%	Error 0%	Error 0%
Empty 0%	Empty 0%	Empty 0%
15634602	42	2
15647311	41	1
15619304	42	8
15632210	25	2
15631201	28	4
15596800	33	1
15623430	34	9
15663489	29	0
15701354	39	1
15737888	43	2

Figure 37: First 10 rows of Age and Tenure Table

Merging of Queries:

- After the queries have been created with each table having its own keys, they were merged into the original table (Churn_Modelling) – which now becomes the fact table simply by virtue of merging these queries into it, using the corresponding columns from each table. (ND: 'Merge Queries' is under the 'Combine' section of 'Home' tab).

Geography.1	Gender.1	IsActive	HasCredit	Churn
Valid 100%	Valid 100%	Valid 100%	Valid 100%	Valid 100%
Error 0%	Error 0%	Error 0%	Error 0%	Error 0%
Empty 0%	Empty 0%	Empty 0%	Empty 0%	Empty 0%
Table	Table	Table	Table	Table
Table	Table	Table	Table	Table
Table	Table	Table	Table	Table
Table	Table	Table	Table	Table
Table	Table	Table	Table	Table

Figure 38: Data view after merging 5 dimensional tables to it

- After merging the tables using the left-outer join system, the matching columns from the original table were then deleted to avoid data redundancy
- Each of the tables were expanded by selecting only the ID columns
- The Age and Tenure columns were also removed from the main table since they were in a separate table (Age and Tenure), then finally removed 'NumOfProducts' as it didn't fit into the storyline of the analytics.

1 ² 3 CustomerID	1 ² 3 GeographyID	1 ² 3 GenderID	1 ² 3 IsActiveID	1 ² 3 HasCreditID	1 ² 3 ChurnID	1 ² 3 CreditScore	\$ Balance	\$ EstimatedSalary
Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%
1	15634602	11	0	1	1	1	619	0.00
2	15632210	13	1	0	1	0	657	171,770.55
3	15647311	12	0	1	0	0	608	83,807.86
4	15596800	13	1	1	1	1	779	158,456.76
5	15621201	12	0	0	1	0	472	0.00
6	15623430	11	1	0	1	0	672	0.00
7	15619304	11	0	0	1	1	502	159,660.80
8	15663489	13	0	0	1	0	633	138,577.34
9	15701354	11	0	0	0	0	699	0.00
10	15737888	12	0	1	1	0	850	125,510.82

Figure 39: First 10 rows of the Fact Table after Pre-processing and Data Cleaning

- After the pre-processing stage, the fact table was reduced to 9 columns: 1 primary key, 5 foreign keys, and 3 numerical columns.

II. DATA MODELLING

The first stage of this section (Creation of Dimensional Tables for Relationships) had already been addressed in the previous section. In this section, we look critically on how this addressed the BI Questions.

The data model at first lacked categorical variables (originally 3 out of 14), there was need to form more categorical variable to have substantial comparative analysis across different dimensions. From Age, Tenure, and Salary – 3 other categorical tables: Age_Group, Tenure_Class, Salary Range were carved out using DAX and M-Language which will be captured in the later sections.

1 ² 3 Age_GroupID	A ^B C Age_Group	A ^B C Age_Bracket	1 ² 3 Lower_Age	1 ² 3 Upper_Age
Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%
1	1001 Teenagers	18-19 Years	18	19
2	1002 Twenties	20-29 Years	20	29
3	1003 Thirties	30-39 Years	30	39
4	1004 Forties	40-49 Years	40	49
5	1005 Fifties	50-59 Years	50	59
6	1006 Sixties & Above	From 60 Years	60	100

A ^B C Tenure_Group	A ^B C Tenure_Bracket	1 ² 3 Lower_Tenure	1 ² 3 Upper_Tenure
Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%
1 New Customers	0-1 Year	0	1
2 Fresh Customers	2-3 Years	2	3
3 Lasted Customers	4-6 Years	4	6
4 Old Customers	7-9 Years	7	9
5 Veteran Customers	From 10 Years	10	20

A ^B C Salary_Range	A ^B C Salary_Bracket	\$ Lower_Salary	\$ Upper_Salary
Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%
1 Low Salary	0-33.33%Max	0.00	66,666.67
2 Mid Salary	33.33-66.67%Max	66,666.67	133,333.33
3 High Salary	66.67-100%Max	133,333.33	200,000.00

Figure 40: Final view of three other categorical tables

- Age_Group was merged to the Age & Tenure table using index column; but this was done by first creating a conditional column named Age_Group based on the ages

1 ² 3 CustomerId	1 ² 3 Age	1 ² 3 Tenure	ABC 123 Age_Group
Valid 100%	Valid 100%	Valid 100%	Valid 100%
Error 0%	Error 0%	Error 0%	Error 0%
Empty 0%	Empty 0%	Empty 0%	Empty 0%
1 15634602	42	2	Forties
2 15632210	25	2	Twenties
3 15647311	41	1	Forties
4 15596800	33	1	Thirties
5 15631201	28	4	Twenties

- Merged using the common column between Age_Group and 'Age & Tenure' tables

1 ² 3 CustomerId	1 ² 3 Age	1 ² 3 Tenure	ABC Age_Group	Age_Group.1
Valid 100%	Valid 100%	Valid 100%	Valid 100%	Valid 100%
Error 0%	Error 0%	Error 0%	Error 0%	Error 0%
Empty 0%	Empty 0%	Empty 0%	Empty 0%	Empty 0%
1 15634602	42	2	Forties	Table
2 15632210	25	2	Twenties	Table
3 15647311	41	1	Forties	Table
4 15596800	33	1	Thirties	Table
5 15631201	28	4	Twenties	Table

- Removed the corresponding column (Age_Group) from the Age & Tenure table and expanded the merged table using left outer join system
- Then reordered and renamed the columns to be like the figure below

1 ² 3 CustomerId	1 ² 3 Age_GroupID	1 ² 3 Age	1 ² 3 Customer_Tenure
Valid 100%	Valid 100%	Valid 100%	Valid 100%
Error 0%	Error 0%	Error 0%	Error 0%
Empty 0%	Empty 0%	Empty 0%	Empty 0%
1 15634602	1004	42	2
2 15647311	1004	41	1
3 15619304	1004	42	8
4 15632210	1002	25	2
5 15631201	1002	28	4

- Tenure_Class and Salary Range were referenced in the model using DAX calculation.

Again, please refer to where Data Source was discussed under this report. A full description of the whole table and their columns have been justified and according to their relationships.

From the dimensions, it's obvious this will give a snowflake model relationship because while other dimensions will be directly attached to the fact table through one-to-many relationships, Age_Group is attached to the Age & Tenure table – thereby causing a bridge connection to the fact table from Age_Group.

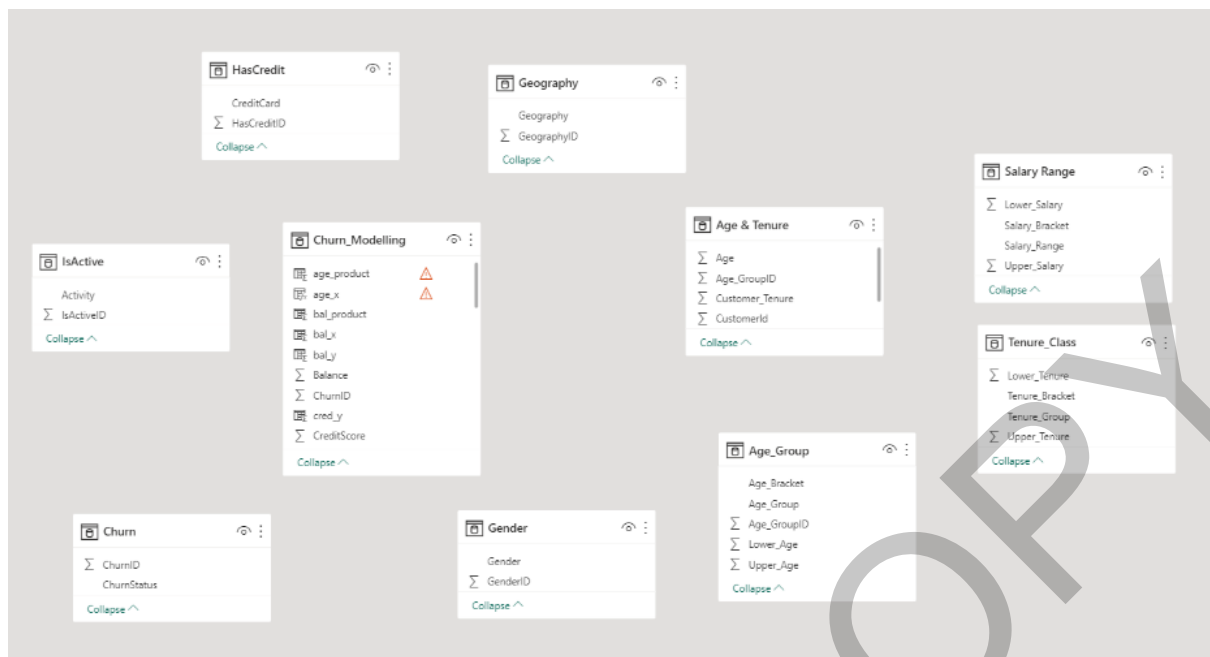


Figure 41: Model View of the Fact Table and Dimensional Table Before Connection

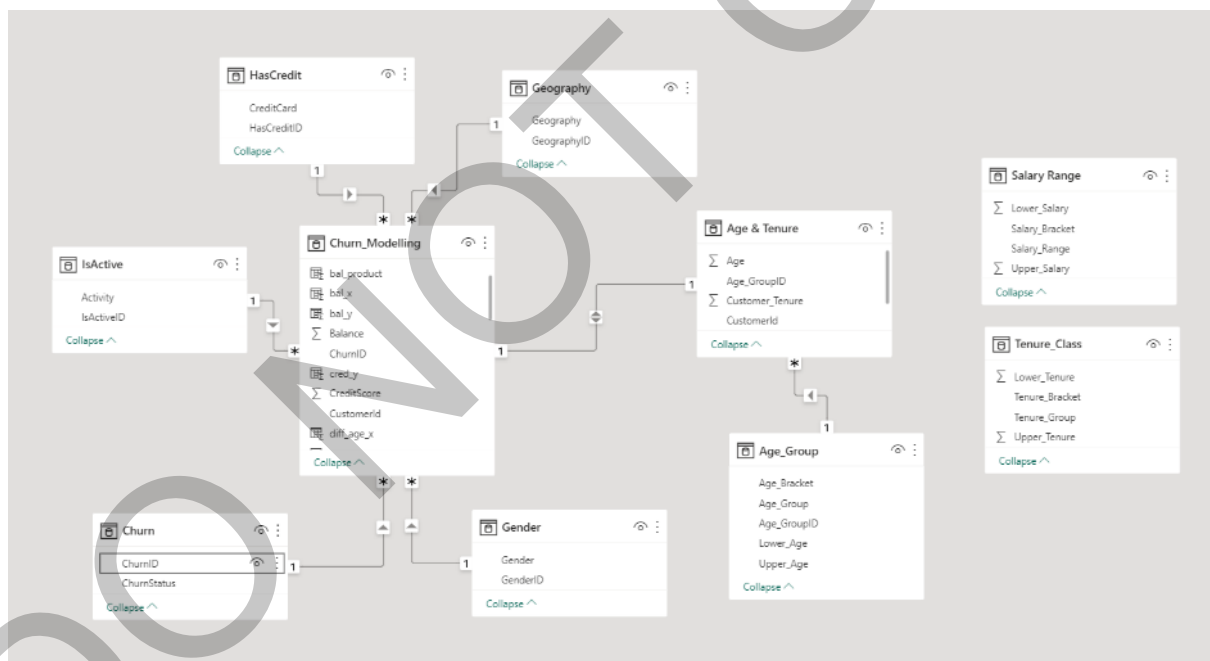


Figure 42: Model View of the Fact Table and Dimensional Table After Connection

III. DAX AND M-LANGUAGE

i) DAX

S/ N	MEASURES	FUNCTIONS	JUSTIFICATION
1	Counts	1) <i>Customers</i> = <code>DISTINCTCOUNT(Churn_Modelling[CustomerId])</code> The DISTINCTCOUNT function was used to count uniquely the number of CustomerId column from the fact table, Churn_Modelling	To quantify the overall count of the customer base for further analysis
2	Conditional Counts	2) <i>Exited</i> = <code>CALCULATE([Customers],Churn[ChurnStatus]="Exited")</code> The CALCULATE function is used to evaluate any expression given to it. It takes two major arguments – the expression and the filter – separated by commas. In the demo, the 'Customers' represents the expression for count while the [ChurnStatus]="Exited" goes to the ChurnStatus column from the Churn table to count all the entries for 'Exited'. This can also take as many conditions (filter) as possible for one single expression, and same as other expressions like sum, average, max, min, etc. 3) <i>Active</i> = <code>CALCULATE([Customers],IsActive[Activity]="Active")</code> 4) <i>Retained</i> = <code>CALCULATE([Customers],Churn[ChurnStatus]="Retained")</code> 5) <i>Has Credit</i> = <code>CALCULATE([Customers],HasCredit[CreditCard]="Yes")</code> 6) <i>Inactive NonZero Acct</i> = <code>CALCULATE([Customers],Churn_Modelling[Balance]>0,IsActive[Activity]="Inactive")</code> Here, two conditions have been evaluated: goes to the balance column from the fact table to count the customerid whose balance is zero and whose Activity column from IsActive table is 'Inactive'. The function evaluates these conditions and count the customerid based on the two filters 7) <i>Active Zero Balance</i> = <code>CALCULATE([Customers],Churn_Modelling[Balance]=0,IsActive[Activity]="Active")</code>	To quantify different kinds of customer based on different dimensions; Exited, Active, Has Credit, Retained, Inactive NonZero Account, and Active Zero Acct. These measures were used to analyze the data better

3	Sum	8) <i>Total Balance</i> = <code>sum(Churn_Modelling[Balance])</code> The SUM function is basically applied to numerical columns – which sums the entries into a total figure 9) <i>Total Salaries</i> = <code>sum(Churn_Modelling[EstimatedSalary])</code>	This was used to quantify the total balance and salaries of customers as one of the key performance indicators that will be used through the data story.
4	Average	10) <i>Average Balance</i> = <code>AVERAGE(Churn_Modelling[Balance])</code> Just like sum, the AVERAGE function is used to calculate average of a given numerical distribution in a column or set of numbers. Here, it calculates the average of the balance column from the fact table 11) <i>Average Age</i> = <code>AVERAGE('Age & Tenure'[Age])</code> 12) <i>Average Salary</i> = <code>AVERAGE(Churn_Modelling[EstimatedSalary])</code> 13) <i>Average Tenure</i> = <code>AVERAGE('Age & Tenure'[Customer_Tenure])</code> 14) <i>Avg. CreditScore</i> = <code>AVERAGE(Churn_Modelling[CreditScore])</code>	From the BI Questions, there was need to calculate these averages as they were used to understand the financial status of the customer according to so many dimensions like ChurnStatus, Activity, Salary_Groups etc
5	Percentage	15) <i>Churn Rate</i> = <code>[Exited]/[Customers]</code> The function can be evaluated by simply taking the proportion of a part to whole (divide) and format the number to a percentage. 16) <i>%Active</i> = <code>[Active]/[Customers]</code> 17) <i>%HasCredit</i> = <code>[Has Credit]/[Customers]</code>	This used to get the impact of customer churn through proportions to summarize the whole analysis

Regression:

In statistical analysis, DAX can be used to evaluate mathematical equations of any form. This is because most statistical equations use some basic mathematical operations like, subtraction, sum, divide, percentage, average to evaluate analysis. In Regression,

The expression can be obtained just like the equation[1] here, where x and y are independent and dependent variables respectively.

$$r_{xy} = \frac{\overline{xy} - \bar{x}\bar{y}}{\sqrt{(\overline{x^2} - \bar{x}^2)(\overline{y^2} - \bar{y}^2)}}.$$

The least squares approach chooses $\hat{\beta}_0$ and $\hat{\beta}_1$ to minimize the RSS. Using some calculus, one can show that the minimizers are

$$\begin{aligned}\hat{\beta}_1 &= \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}, \\ \hat{\beta}_0 &= \bar{y} - \hat{\beta}_1 \bar{x},\end{aligned}\tag{3.4}$$

where $\bar{y} \equiv \frac{1}{n} \sum_{i=1}^n y_i$ and $\bar{x} \equiv \frac{1}{n} \sum_{i=1}^n x_i$ are the sample means. In other words, (3.4) defines the *least squares coefficient estimates* for simple linear regression.

Figure 43: Regression Equation in full context[2]

In this report, 6 linear regression coefficients were obtained using DAX:

- a. Salary vs Balance: y = Balance and x = Salary, beta1 = sal_bal_slope, beta0 = sal_bal_intercept. Thus, y = beta1(x) + beta0

`sal_numerator = SUM(Churn_Modelling[sal_product])`

`sal_denominator = SUM(Churn_Modelling[diff_sal_x_sqrd])`

`sal_bal_slope = [sal_numerator]/[sal_denominator]`

`sal_bal_intercept = AVERAGE(Churn_Modelling[bal_y]) - [sal_bal_slope]*AVERAGE(Churn_Modelling[sal_x])`

sal_product is a calculated column that for the products of the mean difference of x and mean difference of y: `sal_product = Churn_Modelling[diff_sal_x]*Churn_Modelling[diff_bal_y]`

where diff_sal_x is a calculated column for the differences between x (salary) and mean of x (average salary): `diff_sal_x = Churn_Modelling[sal_x] - Churn_Modelling[mean_sal_x]`, `sal_x = Churn_Modelling[EstimatedSalary]`, and `mean_sal_x = AVERAGE(Churn_Modelling[sal_x])`.

Same operations apply to diff_bal_y (for the balance column)

The diff_sal_x_sqrd is square of diff_sal_x: `diff_sal_x_sqrd = Churn_Modelling[diff_sal_x]*Churn_Modelling[diff_sal_x]`

- b. Age vs Balance: y = Balance and x = Age, beta1 = age_bal_slope, beta0 = age_bal_intercept
- c. Tenure vs Balance: y = Balance and x = Tenure, beta1 = ten_bal_slope, beta0 = ten_bal_intercept
- d. Salary vs CreditScore: y = CreditScore and x = Salary, beta1 = sal_cred_slope, beta0 = sal_cred_intercept

- e. Balance vs CreditScore: $y = \text{CreditScore}$ and $x = \text{Balance}$, $\beta_1 = \text{bal_cred_slope}$, $\beta_0 = \text{bal_cred_intercept}$
- f. Tenure vs CreditScore: $y = \text{CreditScore}$ and $x = \text{Tenure}$, $\beta_1 = \text{ten_cred_slope}$, $\beta_0 = \text{ten_cred_intercept}$

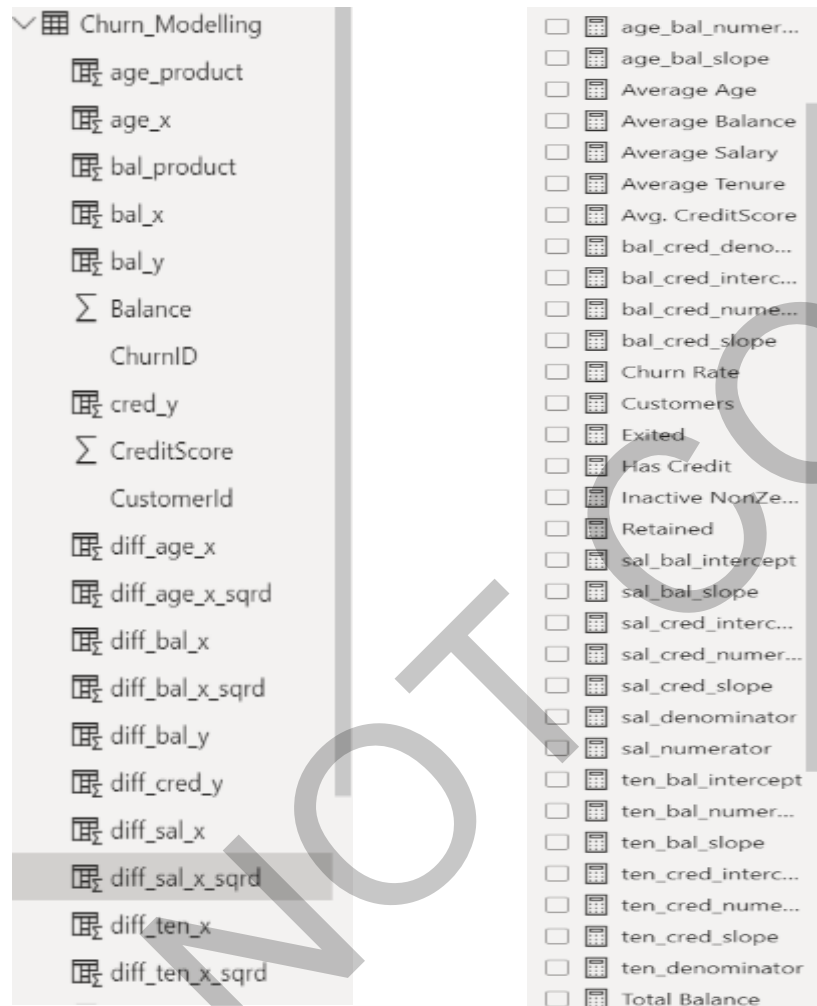


Figure 44: List of Calculated Columns (left) and Measures (right)

Reference Table:

```

1 Salary_Group =
2 VAR class = 'Churn_Modelling'[EstimatedSalary]
3 RETURN
4 CALCULATE(VALUES('Salary_Range'[Salary_Range]),class>'Salary_Range'[Lower_Salary],class<='Salary_Range'[Upper_Salary])

```

Figure 45: Reference Table for Salary_Group using DAX

```

1 Tenure_Group =
2 VAR class = 'Age & Tenure'[Customer_Tenure]
3 RETURN
4 CALCULATE(VALUES(Tenure_Class[Tenure_Group]),class>=Tenure_Class[Lower_Tenure],class<=Tenure_Class[Upper_Tenure])

```

Figure 46:: Reference Table for Tenure Group using DAX

- The VALUES function was used to reference the Salary_Range and Tenure_Group tables that were created using M-Language, and CALCULATE evaluates the conditions of the limits.

ii) M-LANGUAGE:

Age_Group

```
let
Source = Table.FromRecords({
    [Age_Group = "Teenagers", Age_Bracket = "18-19 Years", Lower_Age = 18, Upper_Age=19],
    [Age_Group = "Twenties", Age_Bracket = "20-29 Years", Lower_Age = 20, Upper_Age=29],
    [Age_Group = "Thirties", Age_Bracket = "30-39 Years", Lower_Age = 30, Upper_Age=39],
    [Age_Group = "Forties", Age_Bracket = "40-49 Years", Lower_Age = 40, Upper_Age=49],
    [Age_Group = "Fifties", Age_Bracket = "50-59 Years", Lower_Age = 50, Upper_Age=59],
    [Age_Group = "Sixties & Above", Age_Bracket = "From 60 Years", Lower_Age = 60, Upper_Age=100]
}),
#"Changed Type" = Table.TransformColumnTypes(Source,{{"Age_Group", type text}, {"Lower_Age", Int64.Type}, {"Upper_Age", Int64.Type}, {"Age_GroupID", Int64.Type}},
#"Added Index" = Table.AddIndexColumn("#"Changed Type", "Index", 1001, 1, Int64.Type),
#"Reordered Columns" = Table.ReorderColumns("#"Added Index",{"Index", "Age_Group", "Age_Bracket", "Lower_Age", "Upper_Age"}),
#"Renamed Columns" = Table.RenameColumns("#"Reordered Columns",{{"Index", "Age_GroupID"}})
in
#"Renamed Columns"
```

Figure 47: View of advanced query editor for the Age_Group table (using M-Language)

- Created list of columns and their entries (values)
- Changed the data type
- Added index column that started from 1001 with increment of 1
- Reordered the columns
- Renamed the index column

Tenure_Class

```
let
Source = Table.FromRecords({
    [Tenure_Group = "New Customers", Tenure_Bracket = "0-1 Year", Lower_Tenure = 0, Upper_Tenure=1],
    [Tenure_Group = "Fresh Customers", Tenure_Bracket = "2-3 Years", Lower_Tenure = 2, Upper_Tenure=3],
    [Tenure_Group = "Lasted Customers", Tenure_Bracket = "4-6 Years", Lower_Tenure = 4, Upper_Tenure=6],
    [Tenure_Group = "Old Customers", Tenure_Bracket = "7-9 Years", Lower_Tenure = 7, Upper_Tenure=9],
    [Tenure_Group = "Veteran Customers", Tenure_Bracket = "From 10 Years", Lower_Tenure = 10, Upper_Tenure=20]
}),
#"Changed Type" = Table.TransformColumnTypes(Source,{{"Tenure_Group", type text}, {"Lower_Tenure", Int64.Type}, {"Upper_Tenure", Int64.Type}},
in
#"Changed Type"
```

Figure 48: View of advanced query editor for Tenure_Class

Salary Range

```
let
Source = Table.FromRecords({
    [Salary_Range = "Low Salary", Salary_Bracket = "0-33.33%Max", Lower_Salary = 0, Upper_Salary=66666.67],
    [Salary_Range = "Mid Salary", Salary_Bracket = "33.33-66.67%Max", Lower_Salary = 66666.67, Upper_Salary=133333.33],
    [Salary_Range = "High Salary", Salary_Bracket = "66.67-100%Max", Lower_Salary = 133333.33, Upper_Salary=200000]
}),
#"Changed Type" = Table.TransformColumnTypes(Source,{{"Salary_Range", type text}, {"Lower_Salary", Currency.Type}, {"Upper_Salary", Currency.Type}},
in
#"Changed Type"
```

Figure 49: View of advanced query editor for Salary Range

IV. DASHBOARD DESIGN

Home Page

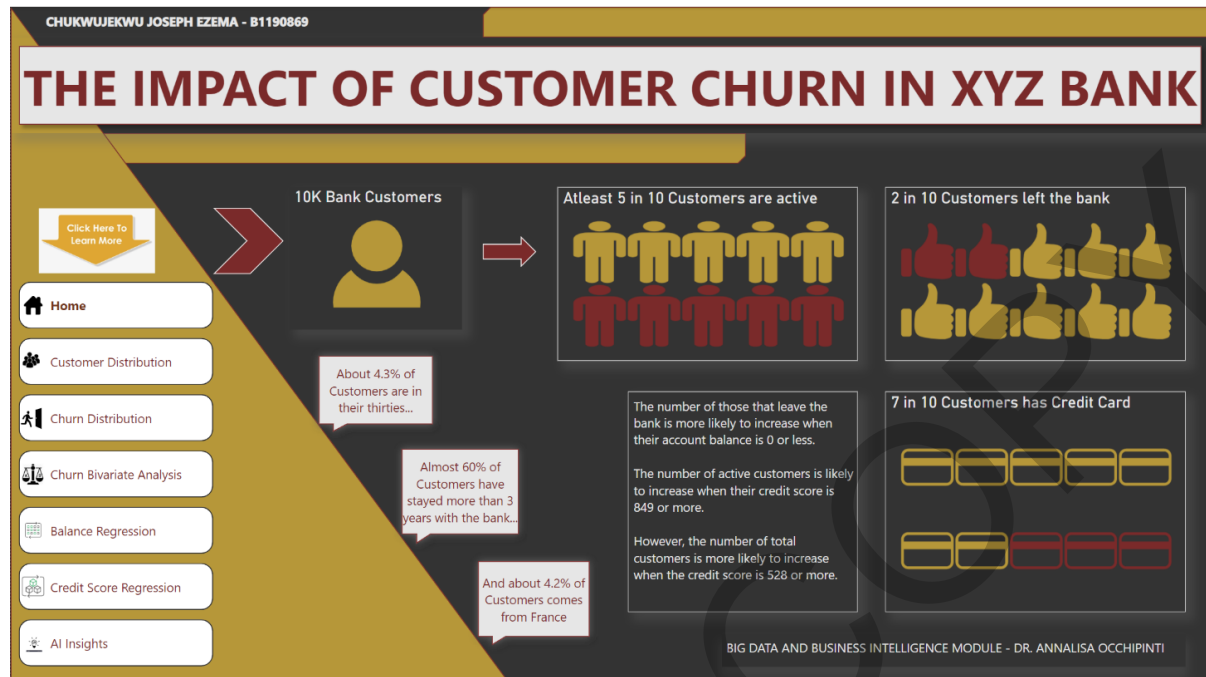


Figure 50: View of Home Dashboard

- The idea of the home was used to capture the content of the report and give a little summary of the work using the infographics, text boxes, and aesthetics (arrows)
- The page buttons were added to give seamless navigation to different pages
- The button icons were added to give clues of what pages were about
- The choice of colour was inspired to give clear contrast with the background colour
- The Title of page reflected the topic of the whole report. However, each page has been titled to reflect the answers to the BI Questions for ease of report flow
- The title of the home page was enlarged to show difference of being the head title

Customer Distribution Page

- This page addresses the first power BI Question – which is the general knowledge of the customer across different dimensions
- It started with total number of customers (10k) since it was the major metric of the page
- Number of exited customers is logically the second measure to give sense of the topic which the impact of those that exited the bank.
- Total Balance is to understand the full financial capacity of the customer base
- Same ideology for the average credit score and average salary
- The summary of the chart types of centres on averages, summation, distribution across dimensions: donut, pie chat, aster plot, bar chart, decomposition tree, table treemap – which are used for distribution analysis

- Use of arrow buttons for navigations across the pages and filters for in-depth analysis

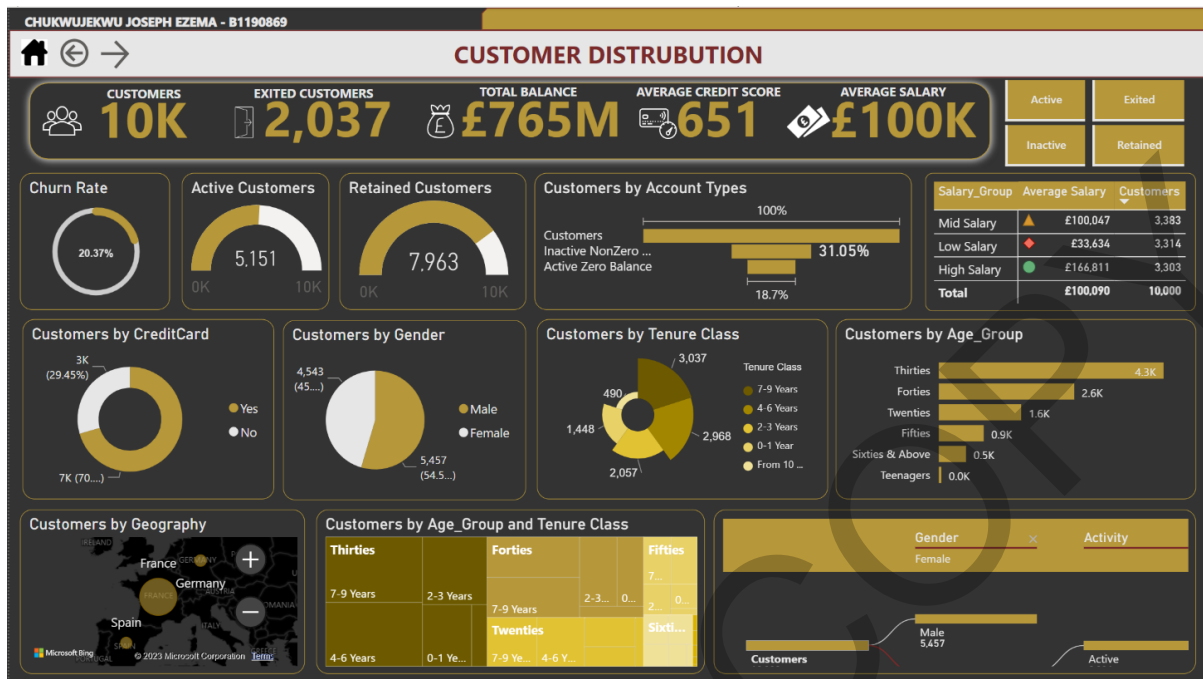


Figure 51: View of Customer Distribution Dashboard

Churn Distribution Page

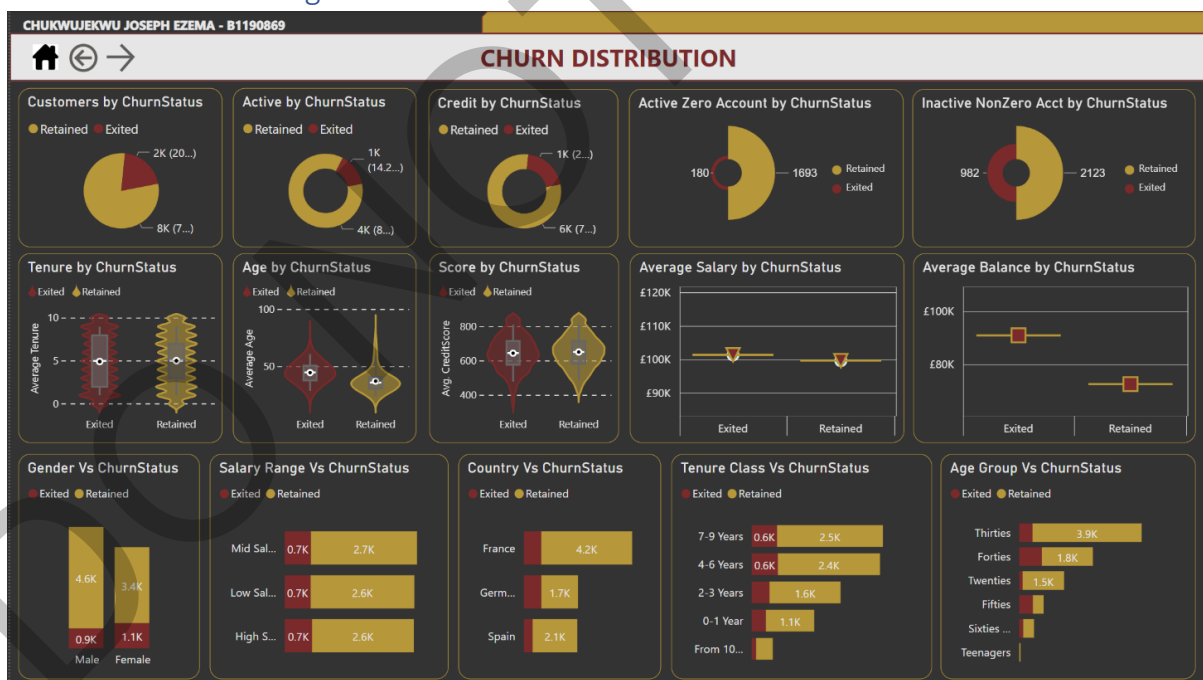


Figure 52: View of Churn Distribution Dashboard

- From the Home Page, maintained the same colour theme (Gold, Wine red, and Grey)
- Addressed BI question 2 for comparison between Exited and Retained customers
- Horizontal alignment of the first top two which in Gestalt principle of similarity, shows the first two line of charts are on univariate analysis of the Churn Status because the BI

questions are all about comparative analysis between the two classes, 'Exited' and 'Retained'

Churn Bivariate Analysis Page

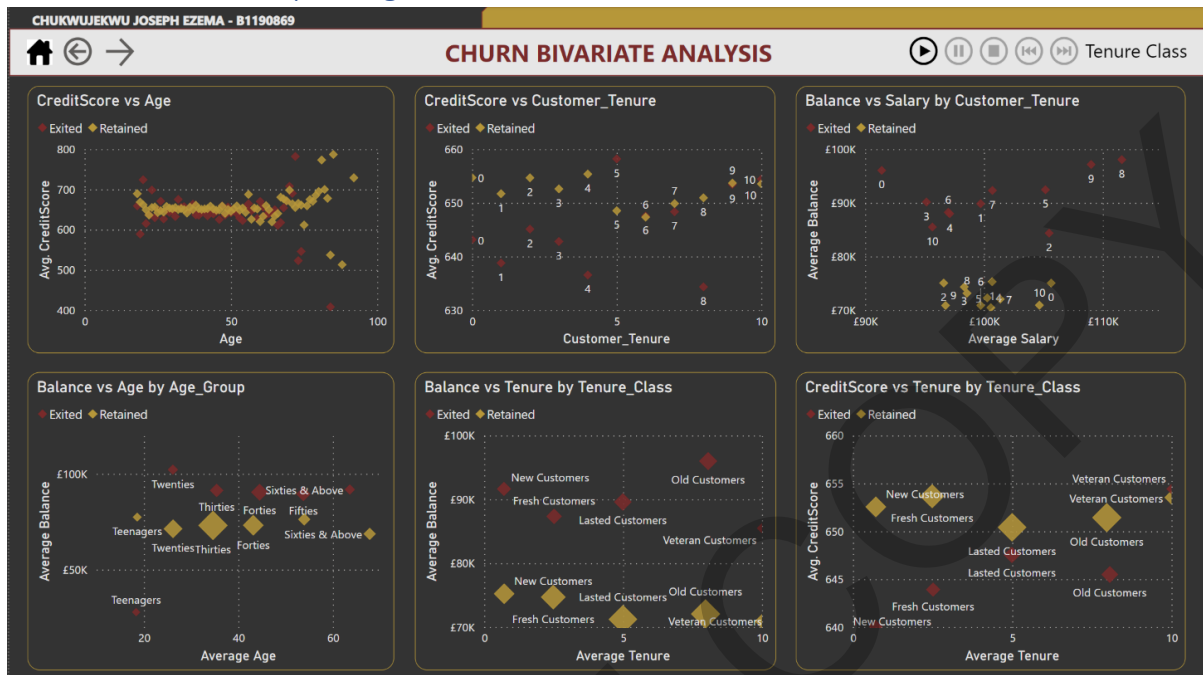


Figure 53: View of Churn Bivariate Analysis Dashboard

- This page explored further the impact of the churn rate by two quantitative variables and a dimension – bivariate analysis of the target class (Churn Status).
- Added a little animation using the Tenure Class filter in a play axis across the analysis

Balance Regression Page

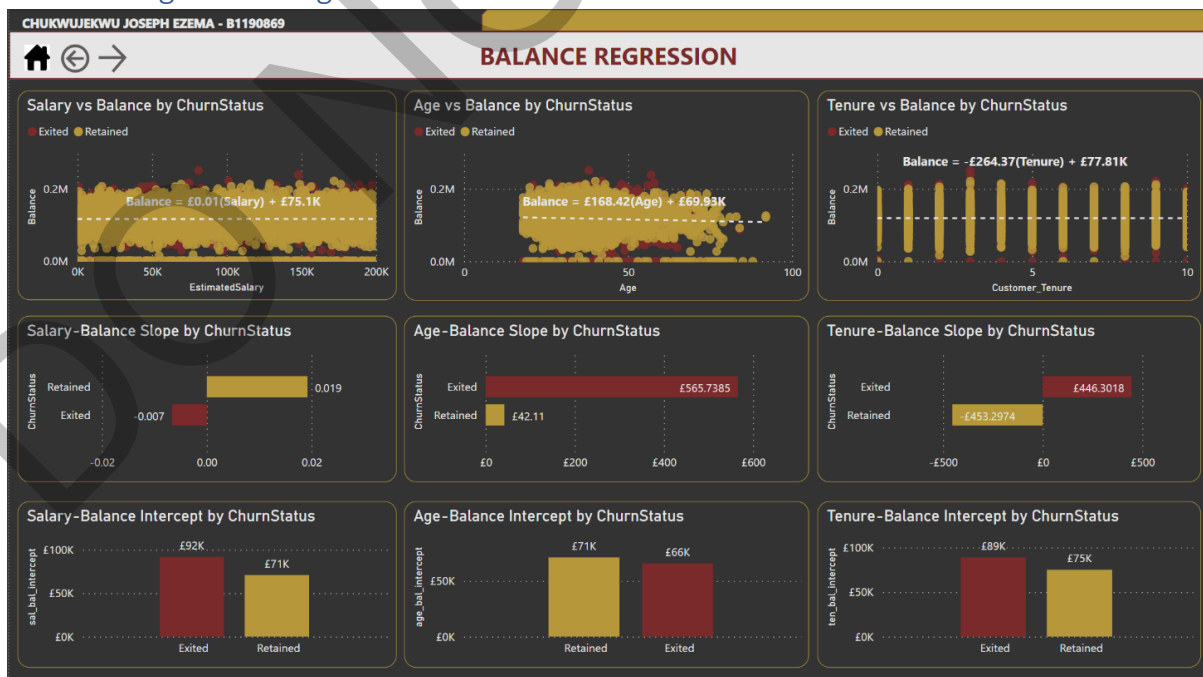


Figure 54: View of Balance Regression Dashboard

- From BI question 3 and 4, this dashboard addressed the question of correlation

Credit Score Regression Page

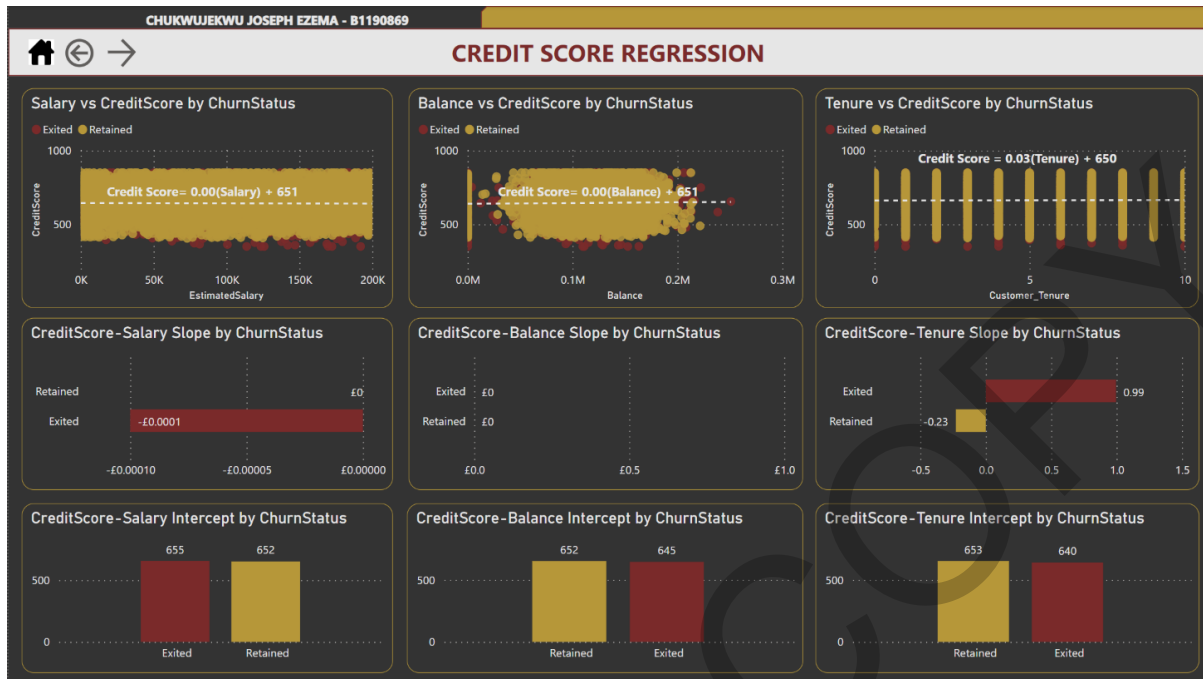


Figure 55: View of Credit Score Regression Dashboard

AI Insights Page



Figure 56: View of the AI Insights Dashboard

- Explored AI-enabled charts like Key Influencers and Decomposition tree to be able to answer the BI question 5 using insights to drive results

V. REFERENCE

- [1] Challenge – Regression Coefficient (Week 6 Lab Session)
- [2] Tutorial 3 – Power BI Workbook – Advanced Maths with DAX (Week 5 Lab Session)
- [3] 5 WAYS to use BOOKMARKS in Power BI [https://youtu.be/ HTF7Ph7Egc](https://youtu.be/HTF7Ph7Egc)
- [4] 5 DESIGN TRICKS that Make EVERY Power BI Report Look GREAT!
<https://youtu.be/v6fP8gyCLLc>

VI. SELF-ASSESSMENT TABLE

Report Section	Description	Grade your work from 0 to 100
Report Structure	The report is well-written, and it contains all the relevant sections	95
Data Pre-processing and Data Modelling	Many pre-processing steps have been applied. The data model is well-structured	90
Dax and M language	Both DAX and M Language have been extensively used in the report	95
Dashboard Design	The dashboard contains a variety of charts, including advanced ones not covered in the module.	90
Average		Add below the average of the four cells above: 92.5